



The **AiEdge** ACADEMY

The Application Layer



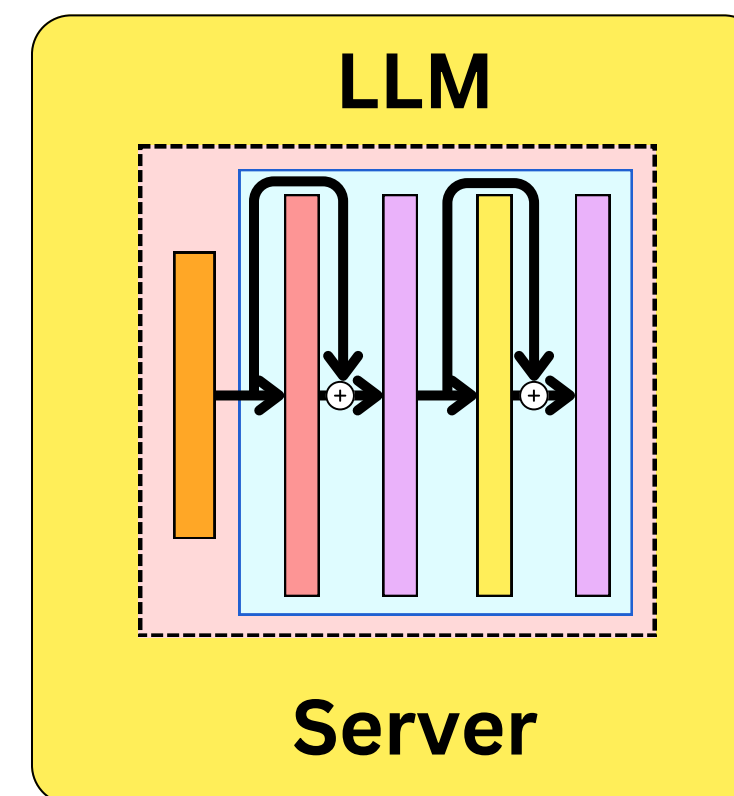
The Application Layer



User



Client

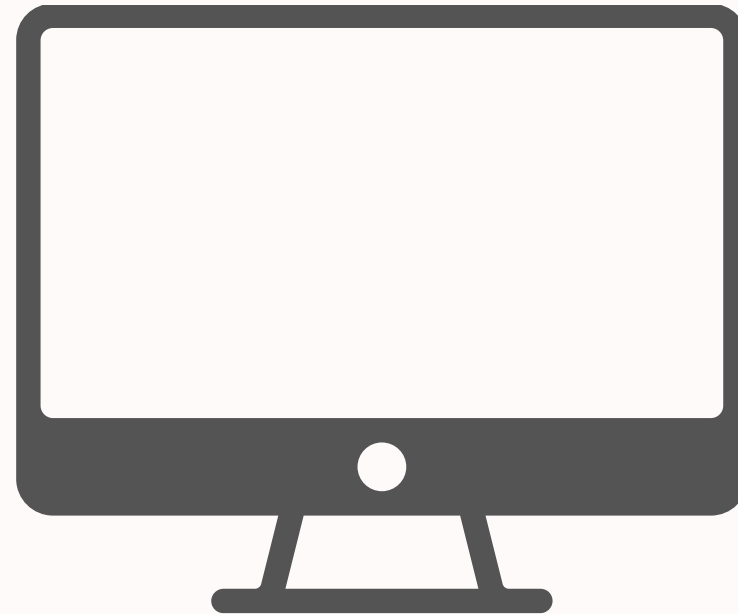




The Application Layer



User



Client

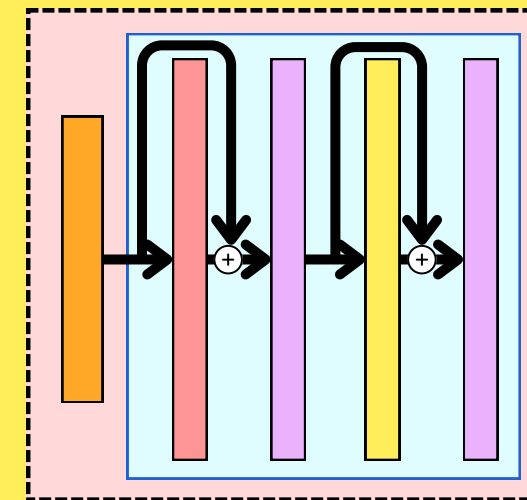
Request



Response



LLM



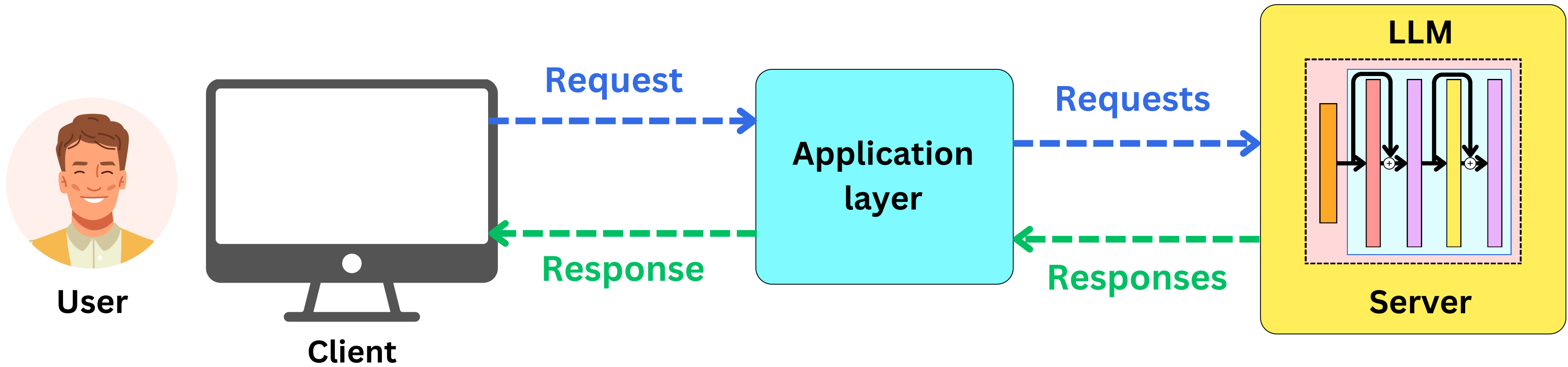
Server



Very limiting implementation



The Application Layer

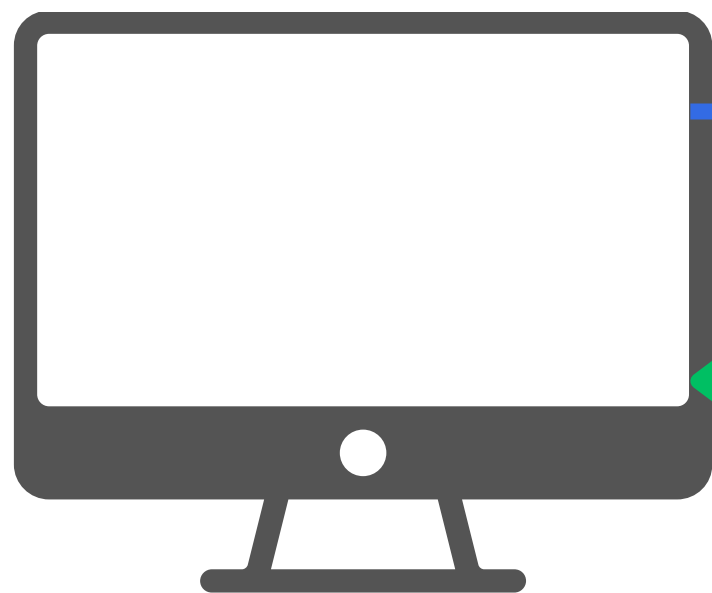




The Application Layer



User



Client

Request



Response



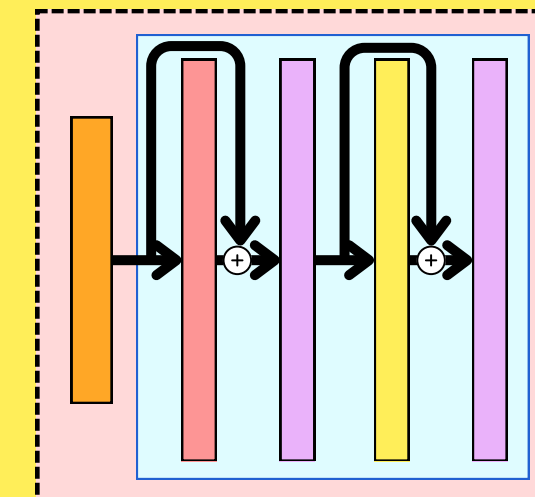
Application
layer

Multiple LLM
requests
Requests



Responses

LLM



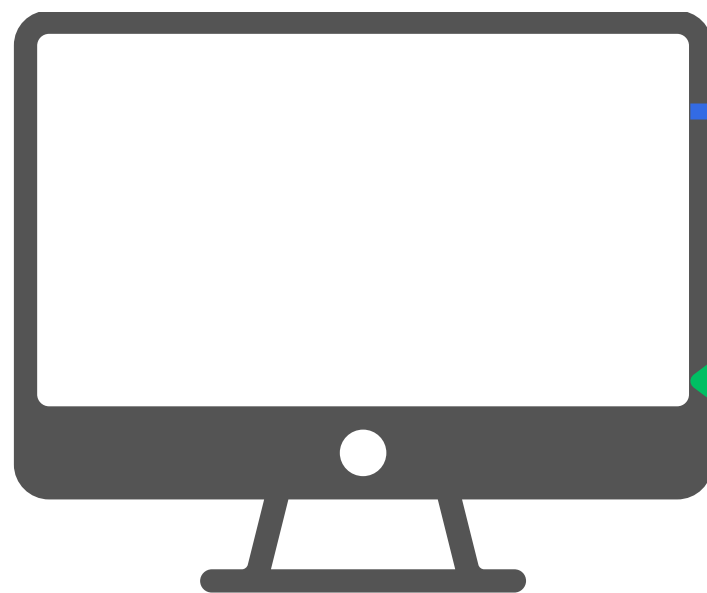
Server



The Application Layer



User



Client

Request

Response

Application
layer

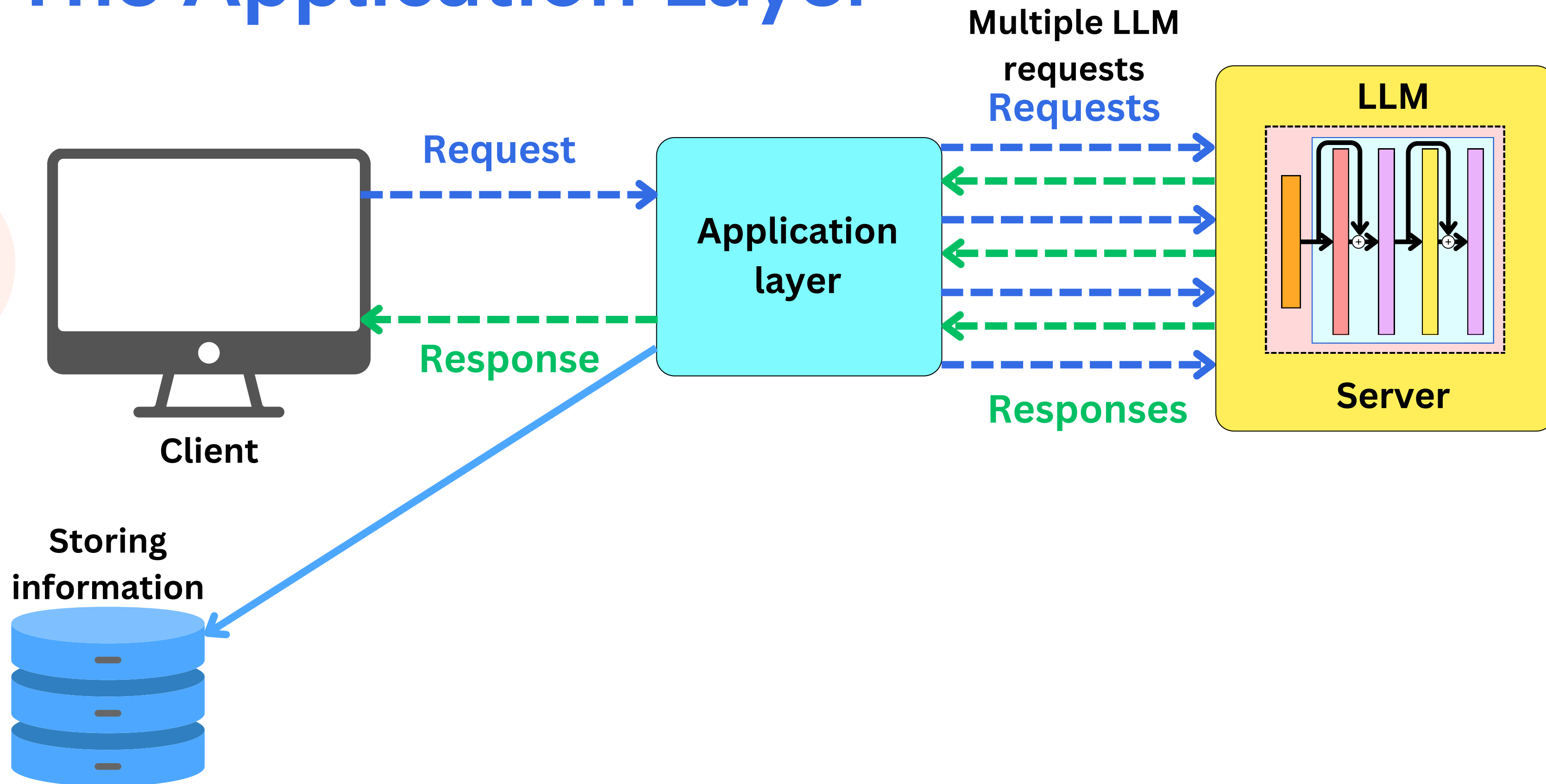
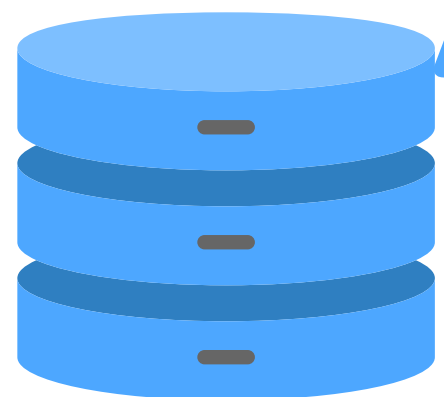
Multiple LLM
requests
Requests

Responses

LLM

Server

Storing
information

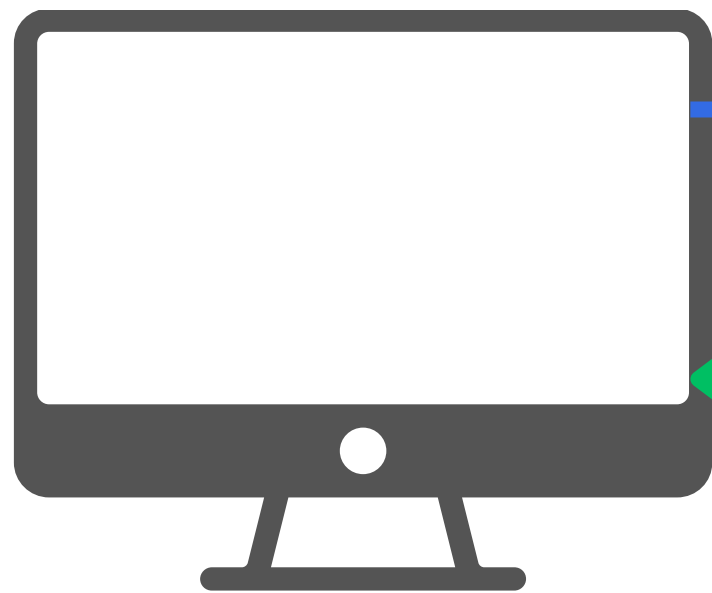




The Application Layer



User



Client

Request

Response

Application
layer

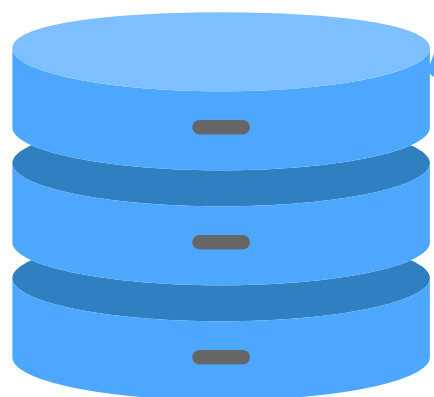
Multiple LLM
requests
Requests

Responses

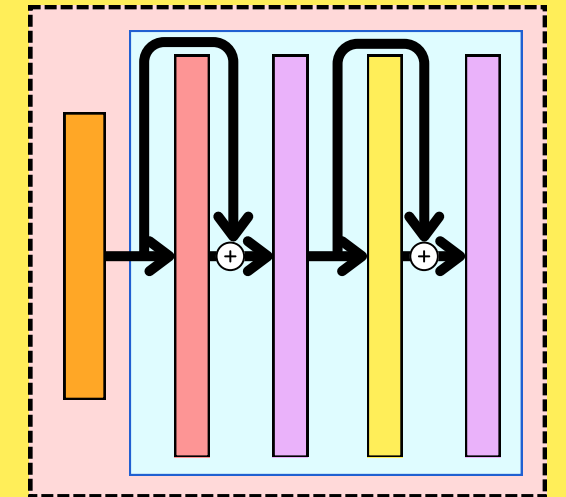
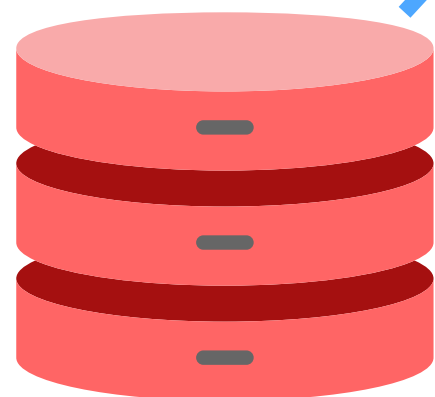
LLM

Server

Storing
information

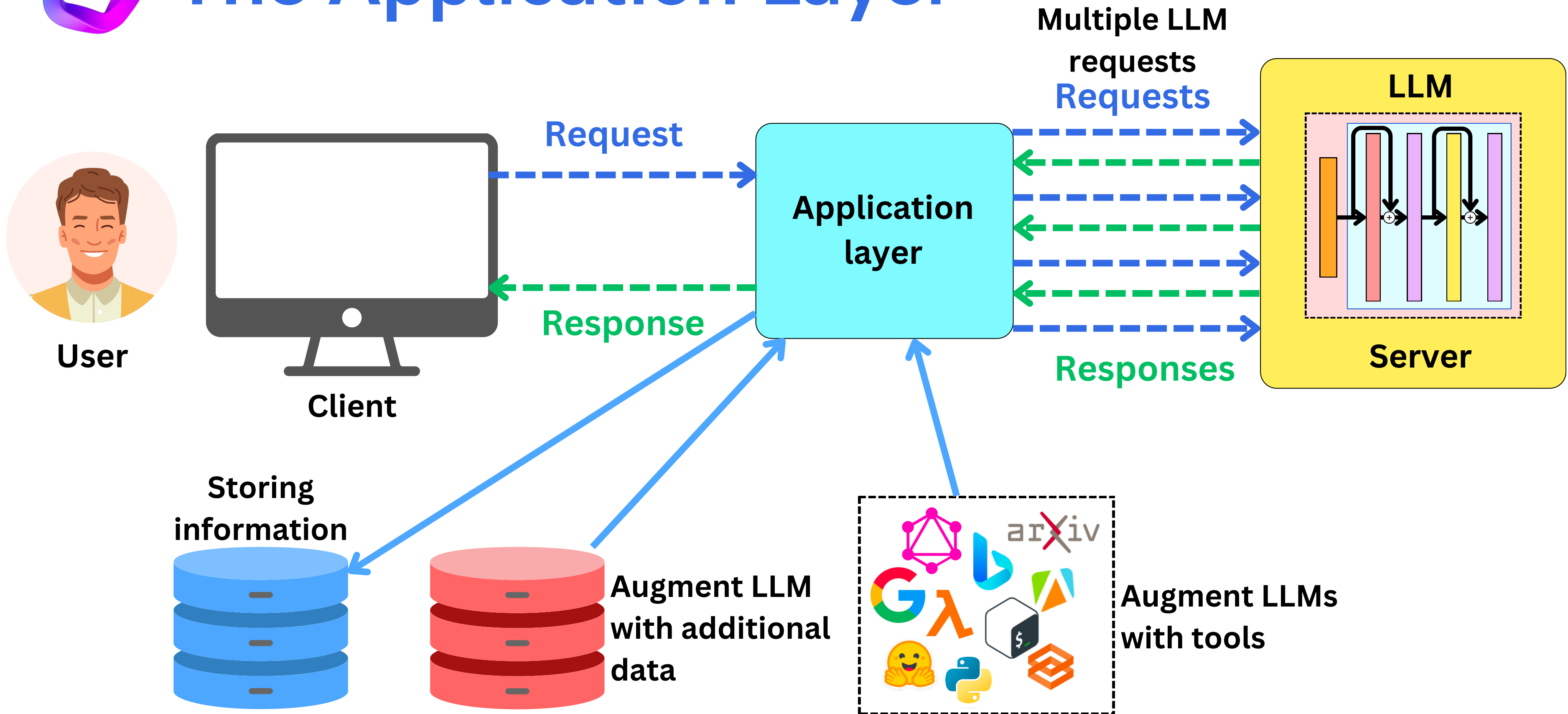


Augment LLM
with additional
data



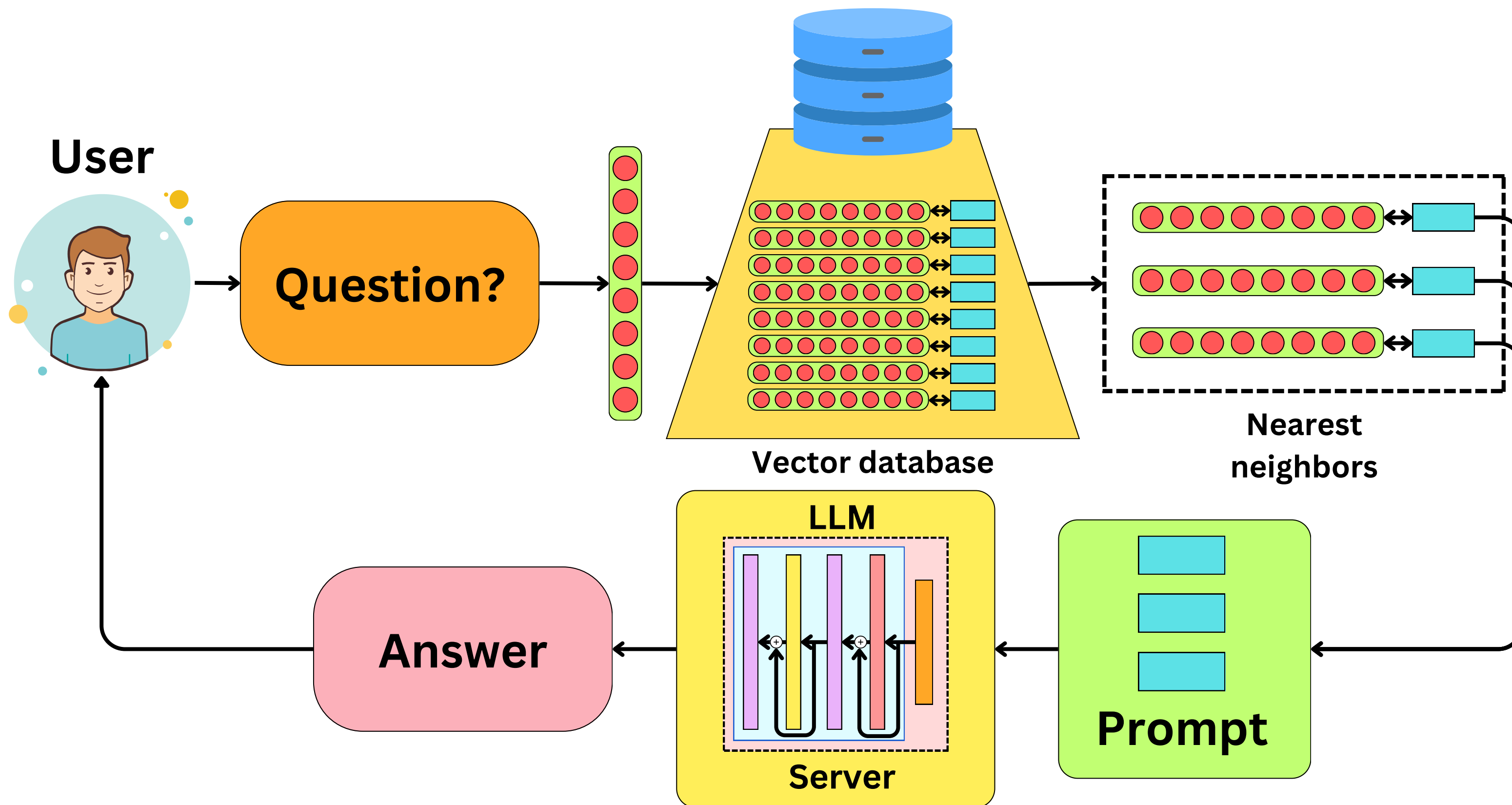


The Application Layer



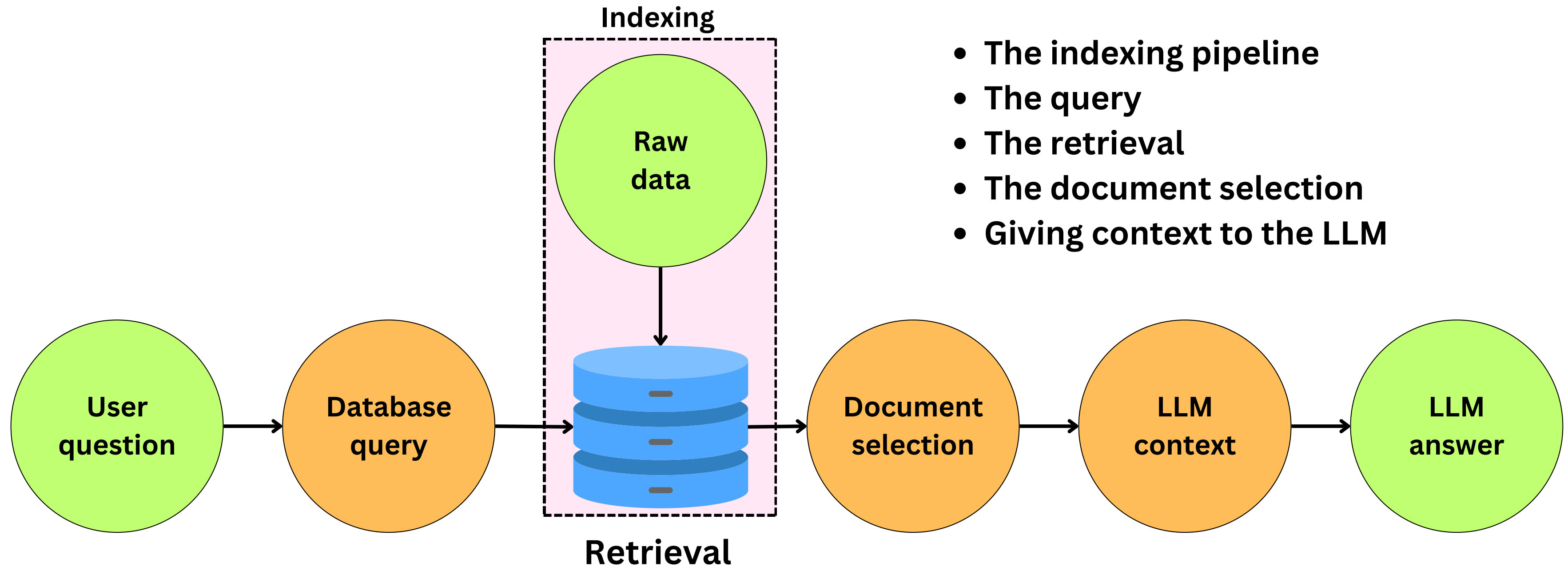


Retrieval Augmented Generation





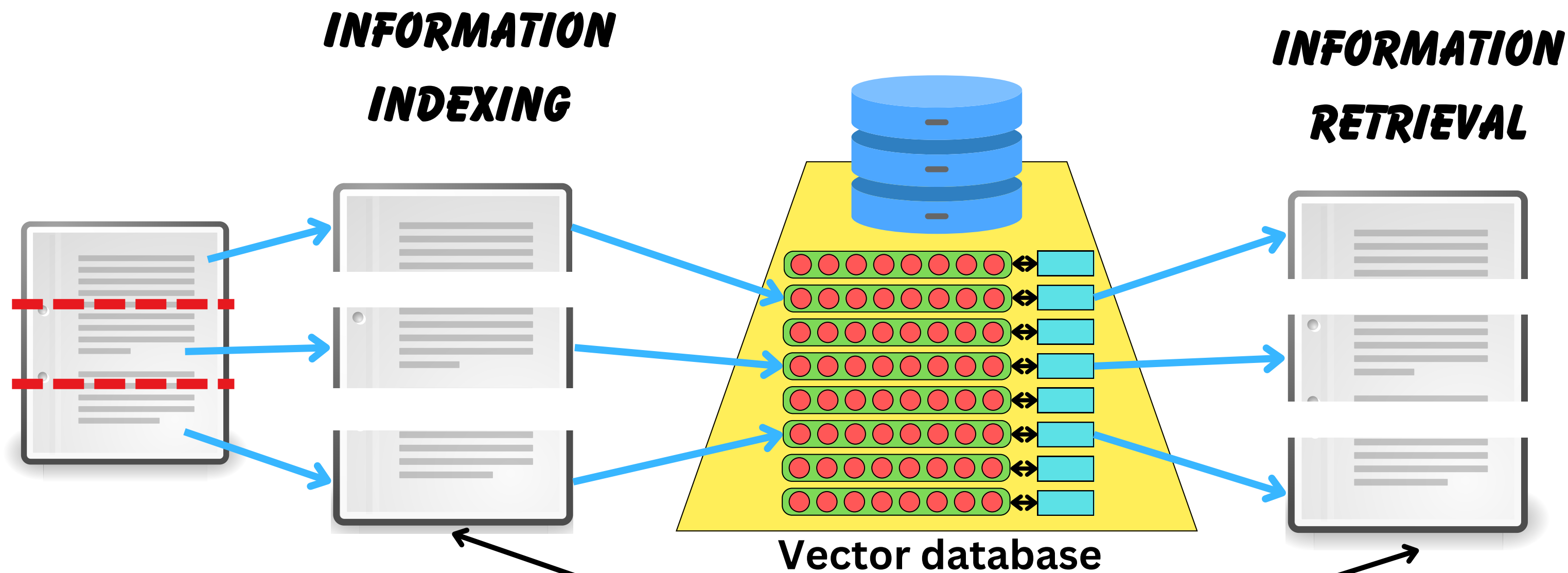
The RAG components





Optimizing the Indexing Pipeline

Typical approach

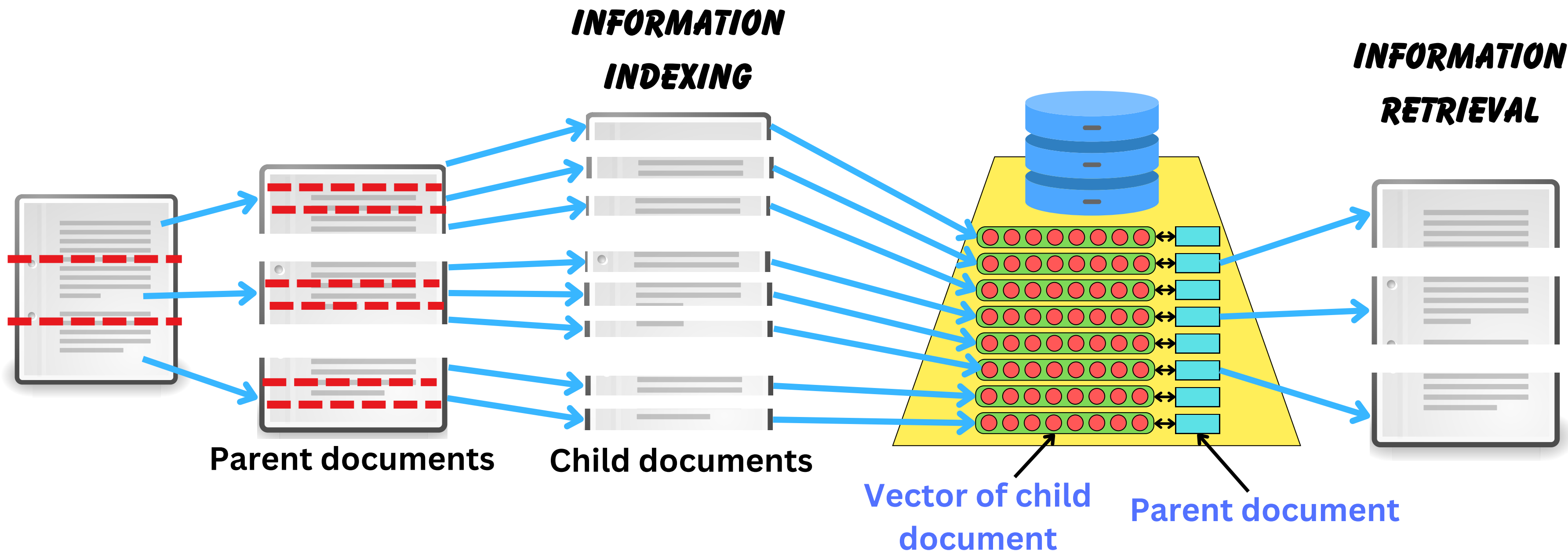


Typical RAG: Same data used
for indexing and retrieval



Optimizing the Indexing Pipeline

Indexing by small data chunks

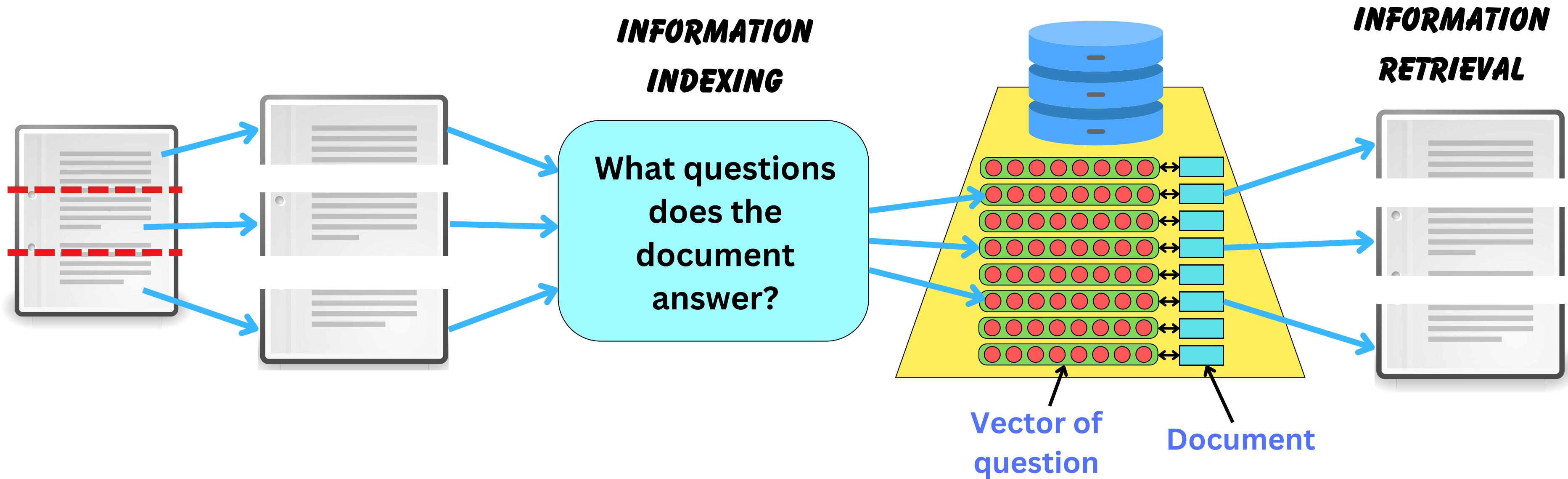


The parent documents are indexed by a vector representation of their child documents



Optimizing the Indexing Pipeline

Indexing by the questions the document answers

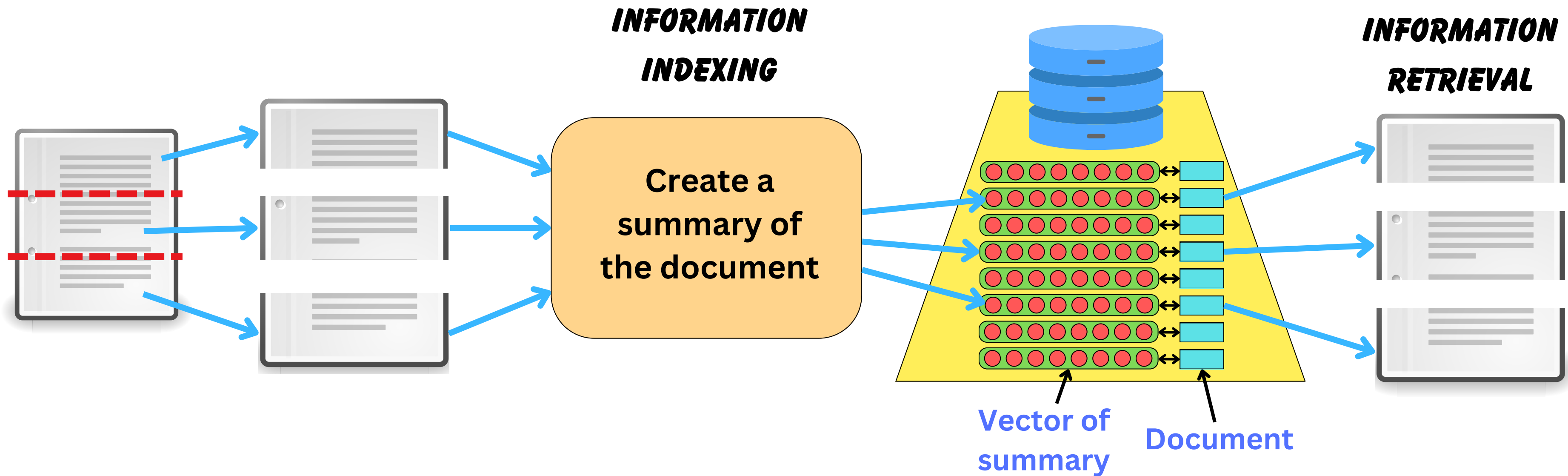


The documents are indexed by a vector representation of the possible questions the documents answer



Optimizing the Indexing Pipeline

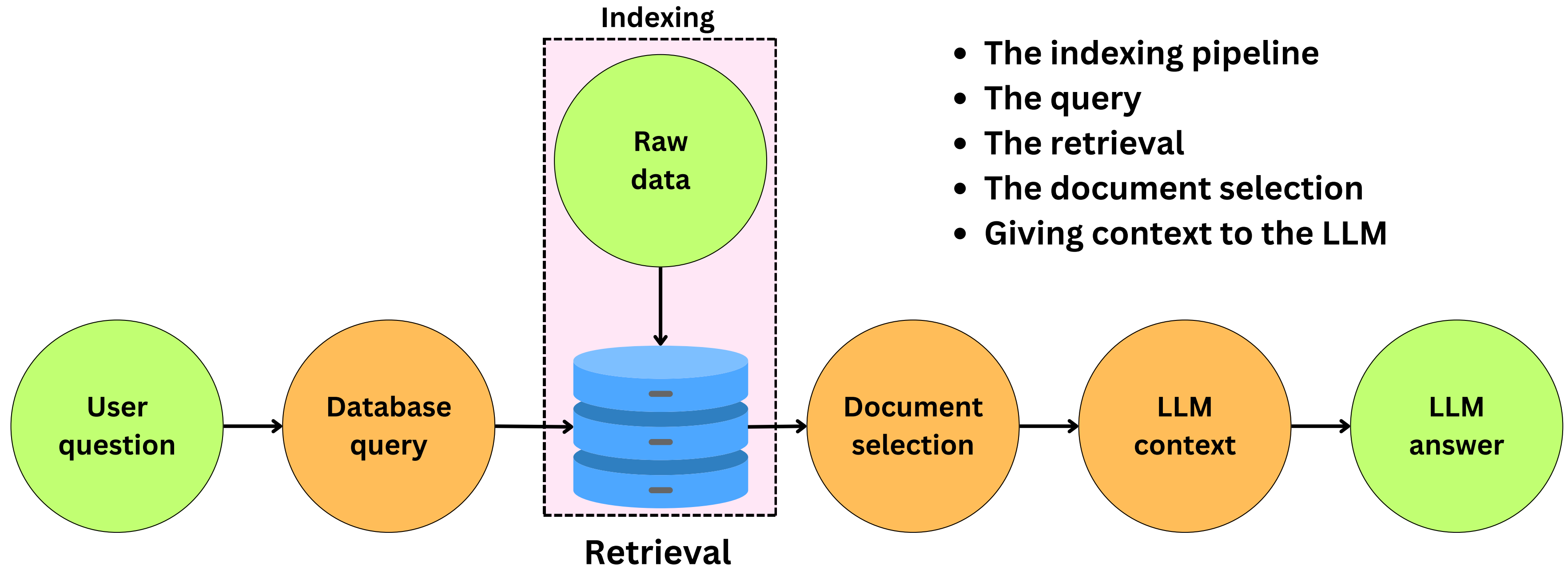
Indexing by the summary of the document



The documents are indexed by a vector representation of their summaries



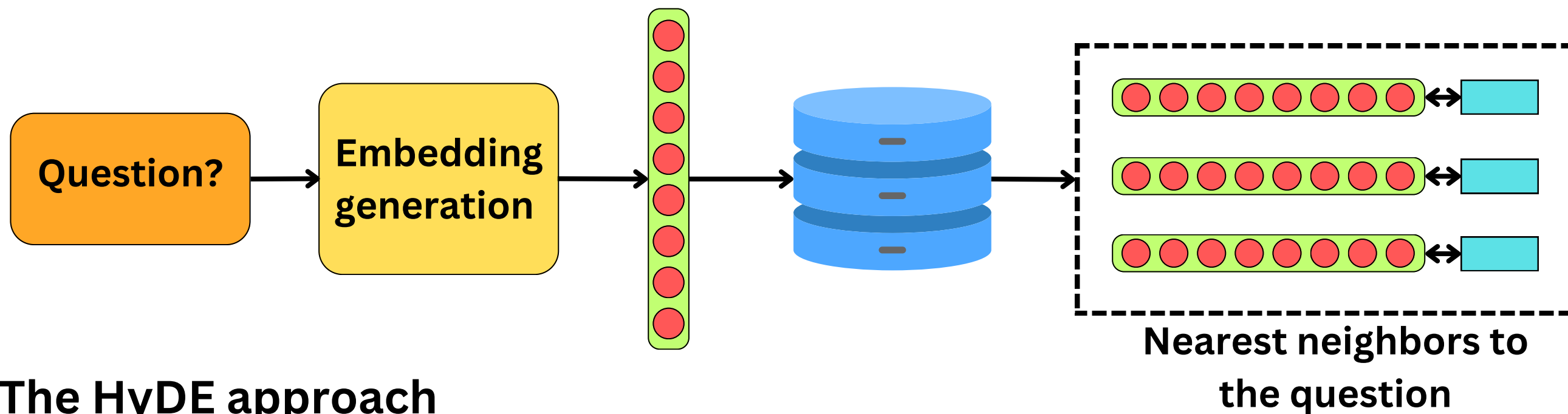
The RAG components



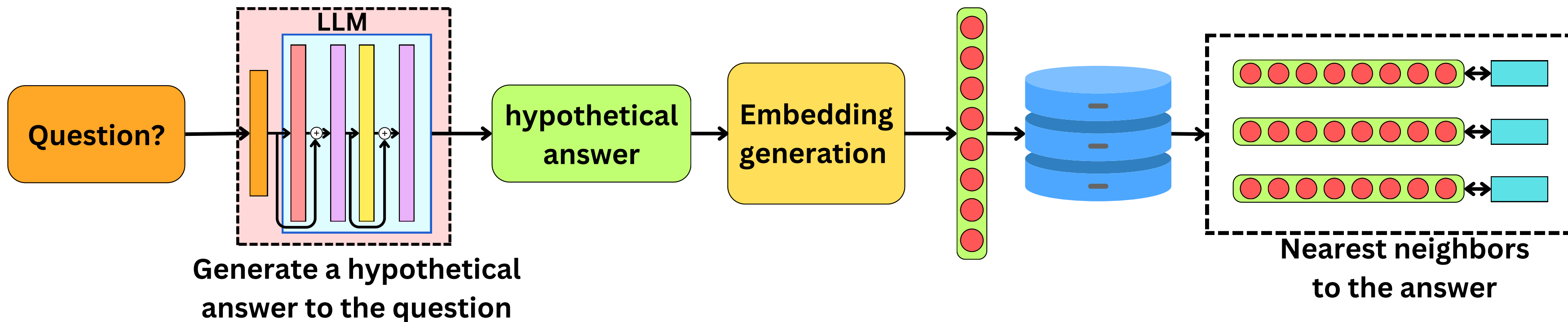


Optimizing the Query

The Typical RAG approach



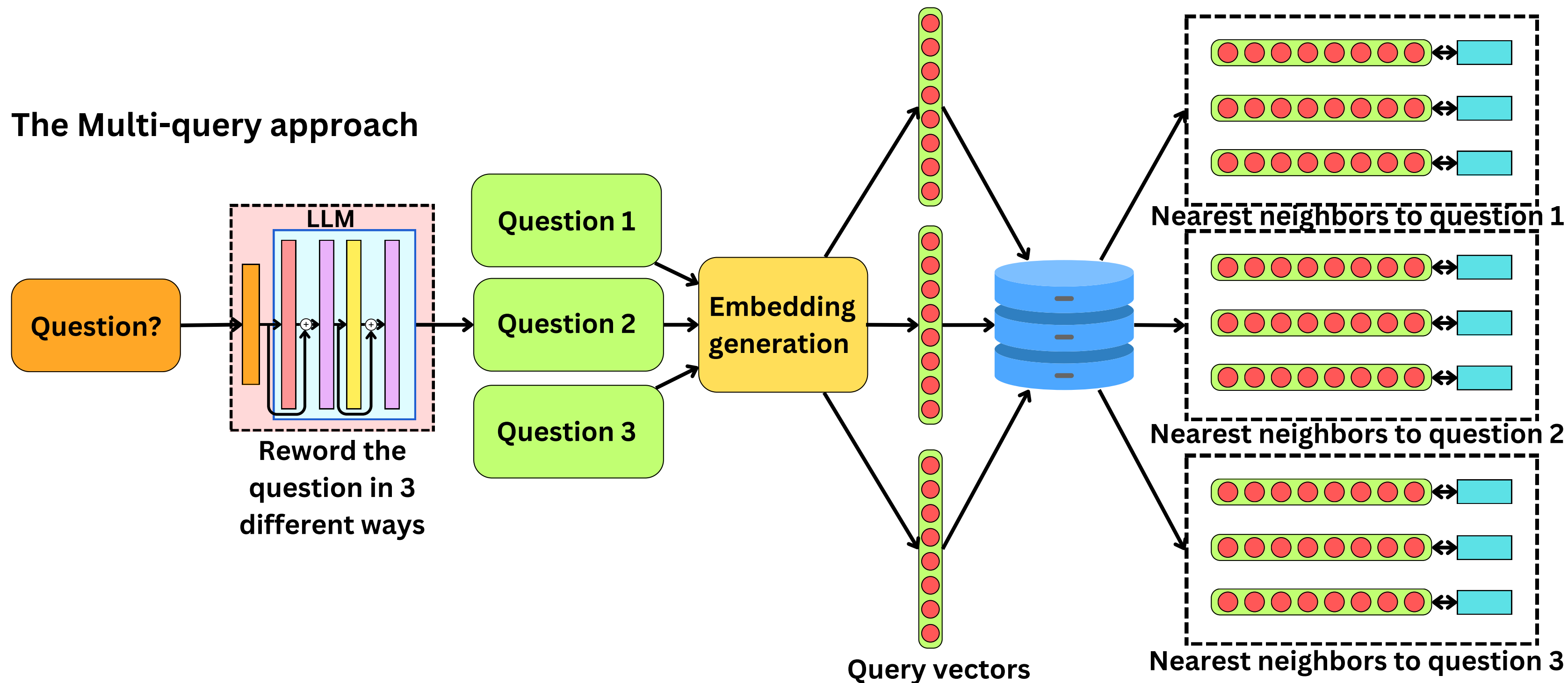
The HyDE approach





Optimizing the Query

The Multi-query approach

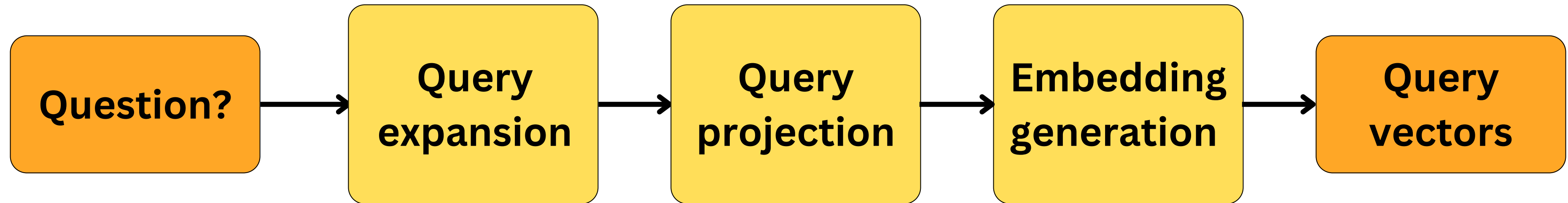




Optimizing the Query

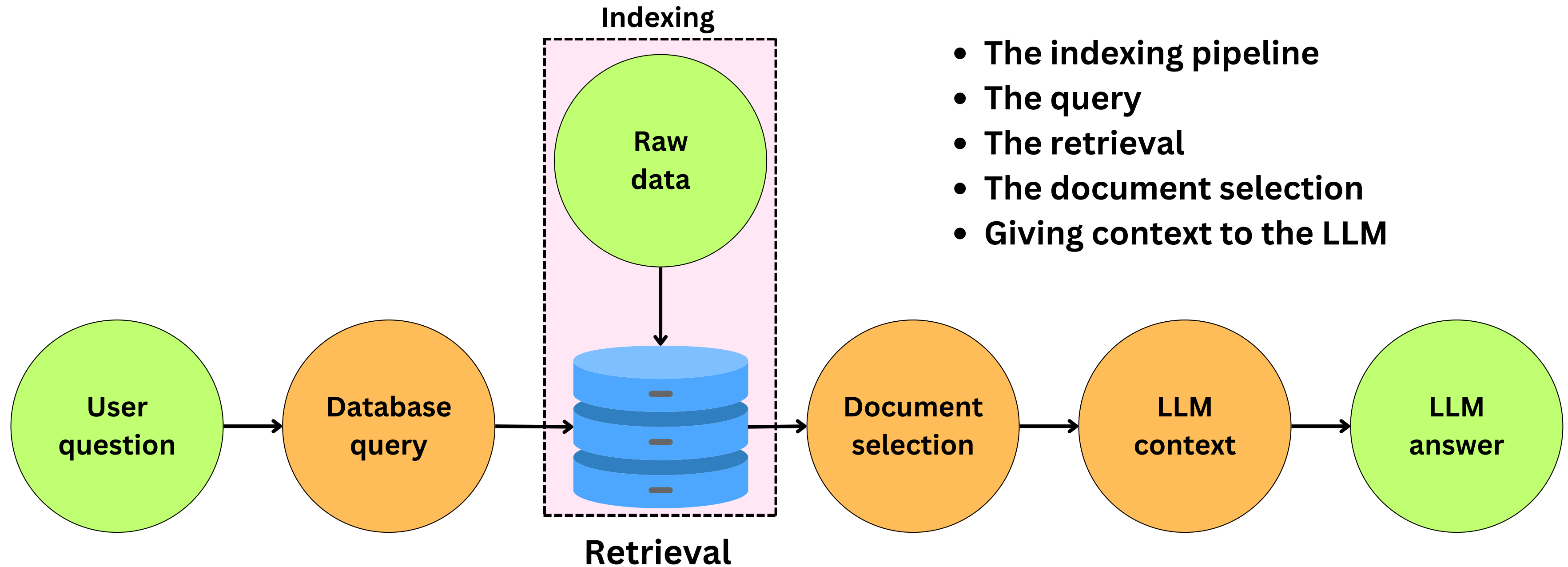
Combining approaches

The Query pipeline





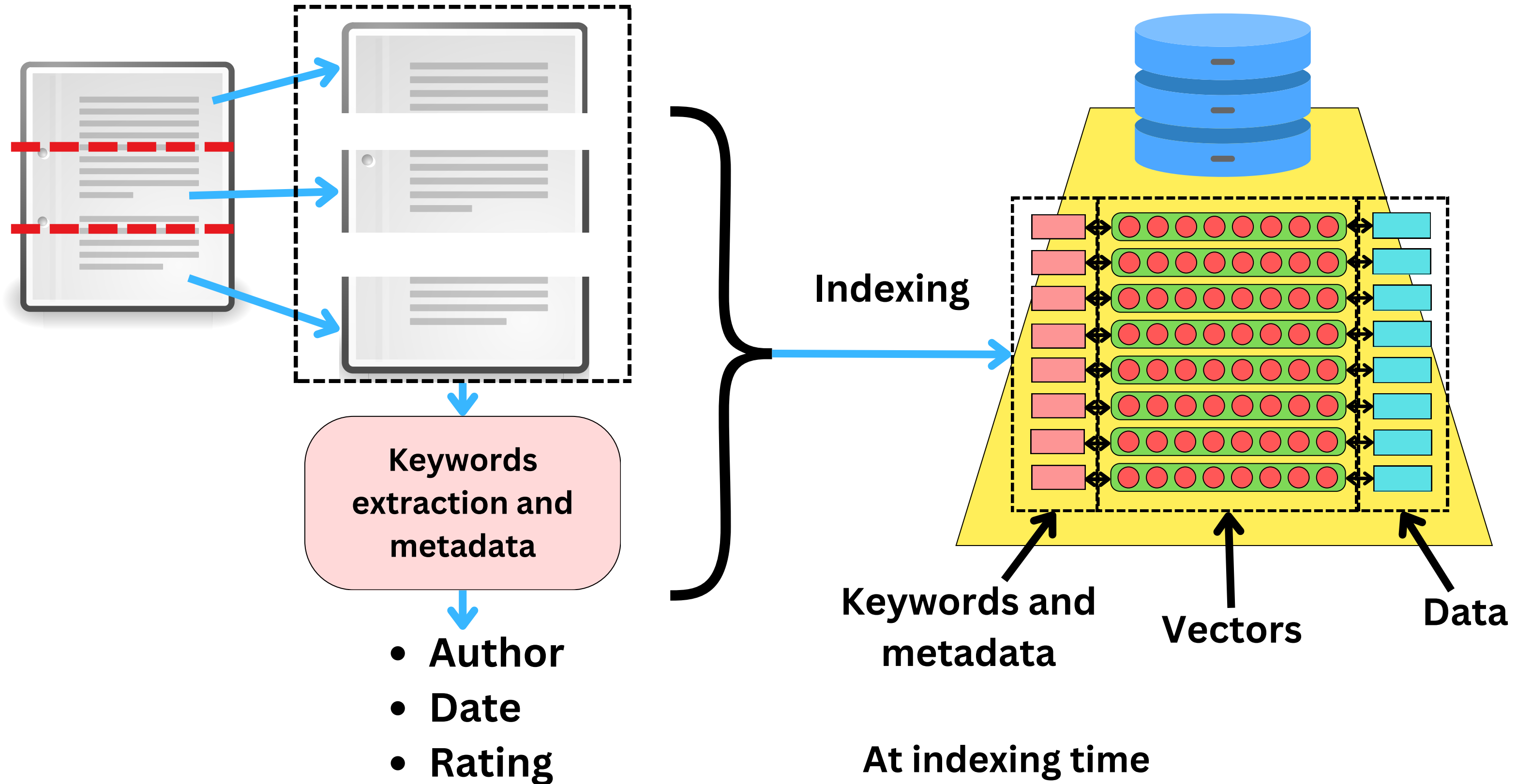
The RAG components





Retrieval optimization

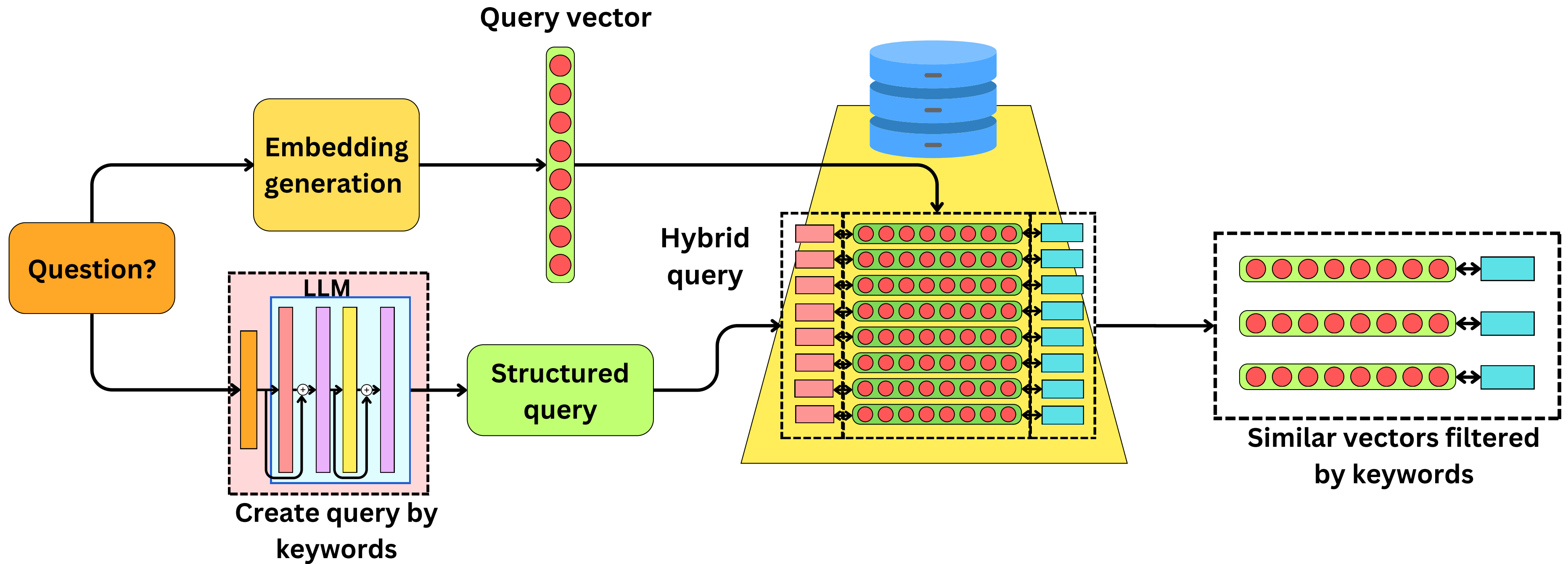
Self-Querying





Retrieval optimization

Self-Querying



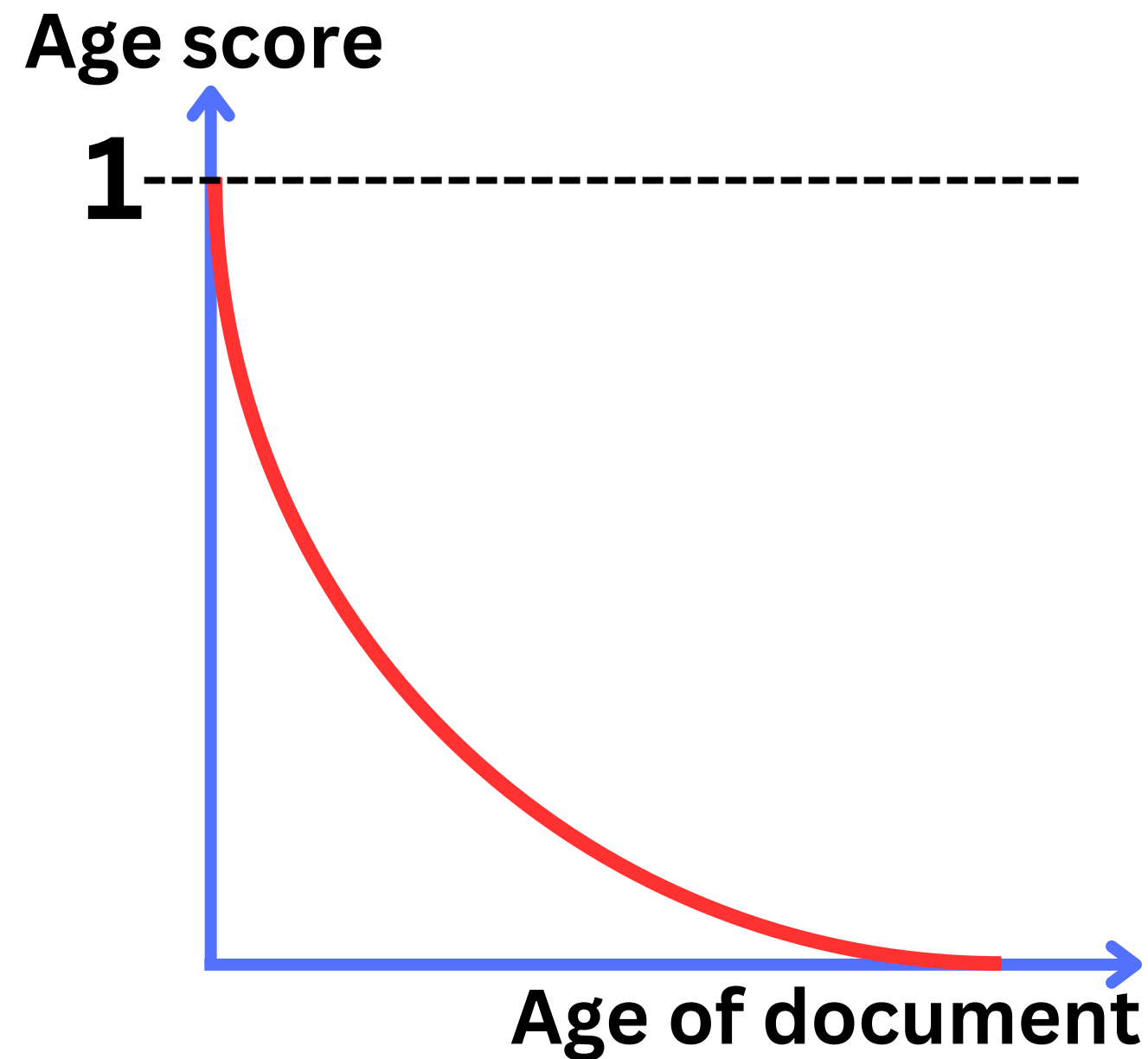
At retrieval time



Retrieval optimization

Time-weighted retriever

$$\text{Retrieval score} = \text{Semantic similarity} + r^{\text{Age of document}}$$

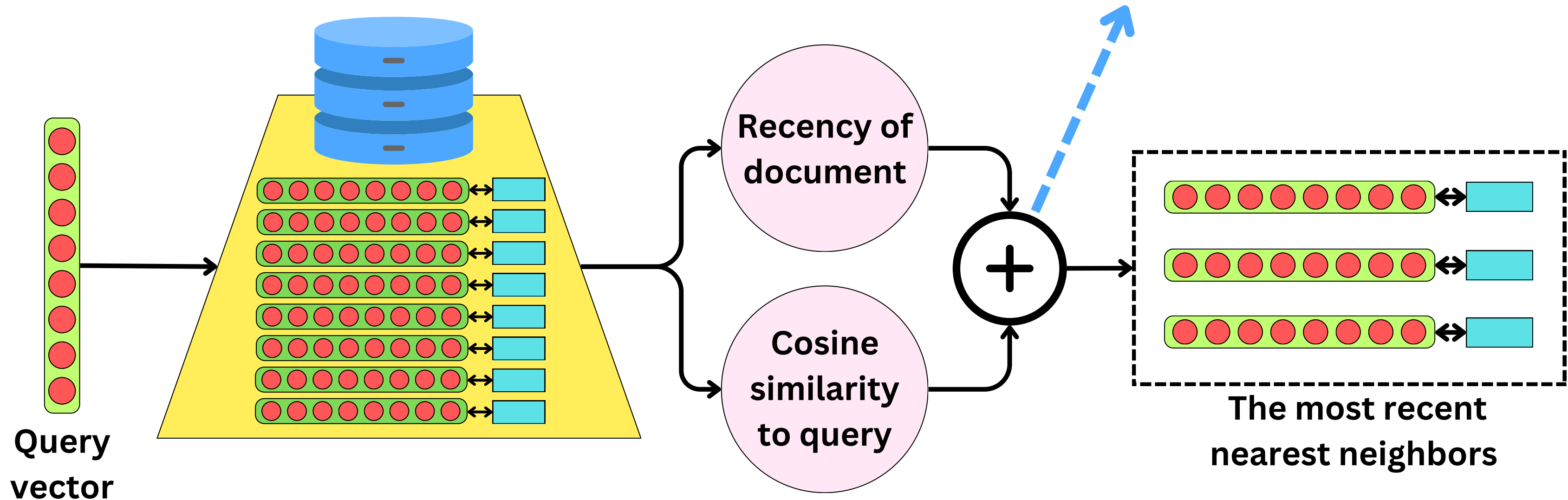




Retrieval optimization

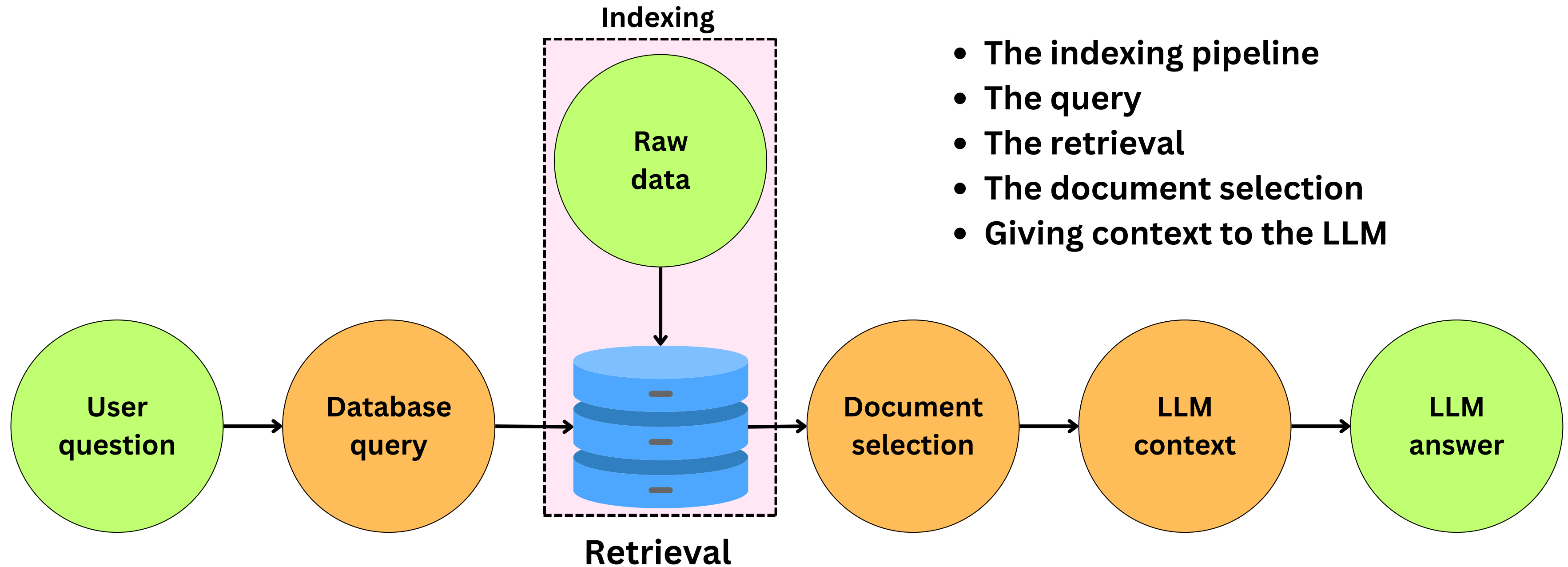
Time-weighted retriever

$$\text{Retrieval score} = \text{Semantic similarity} + r^{\text{Age of document}}$$





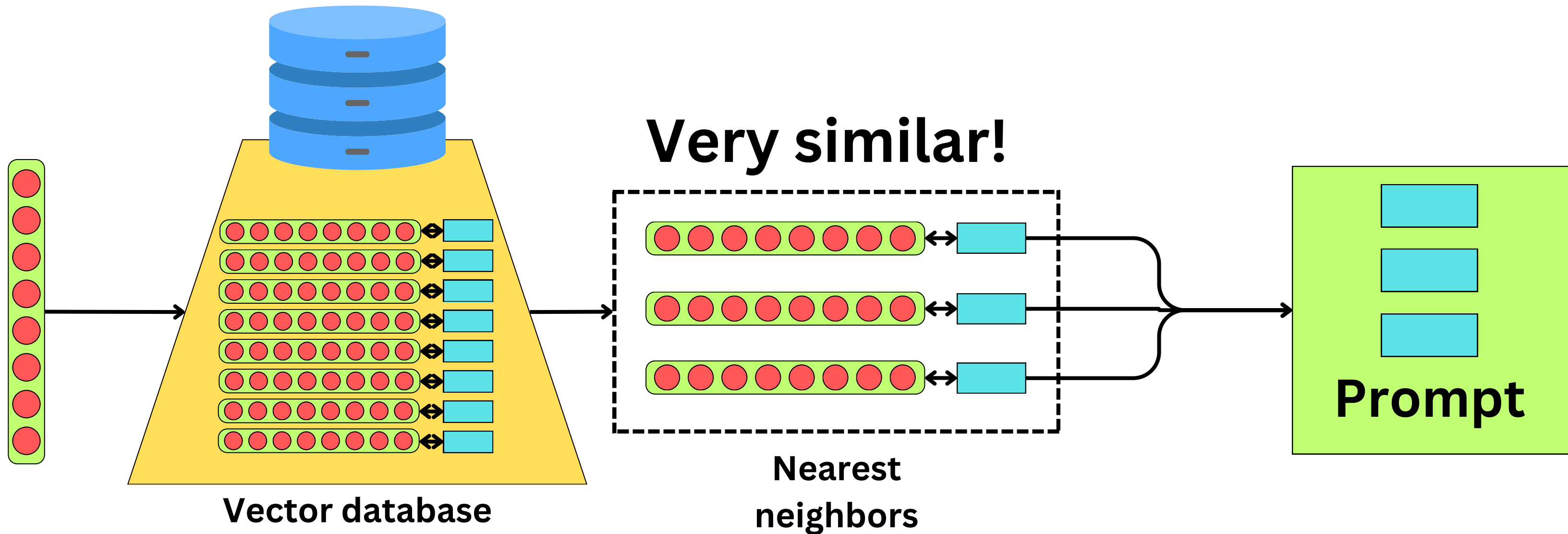
The RAG components





Document selection optimization

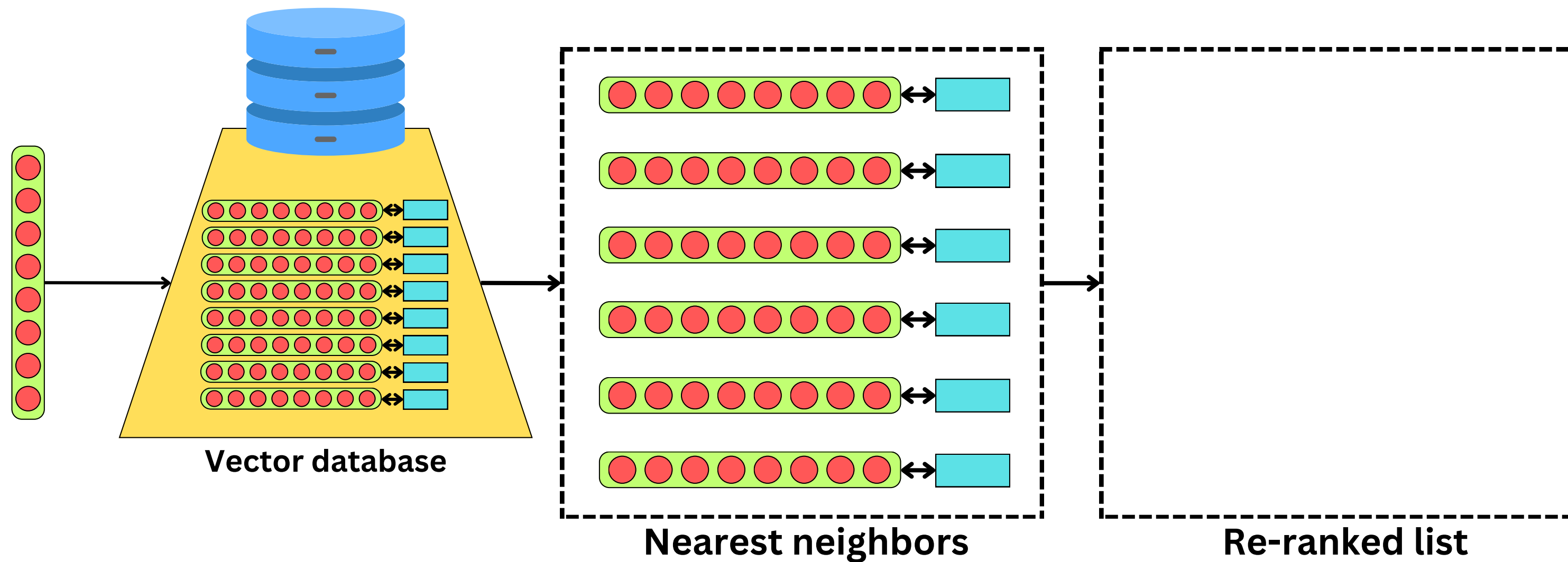
Maximal Marginal Relevance





Document selection optimization

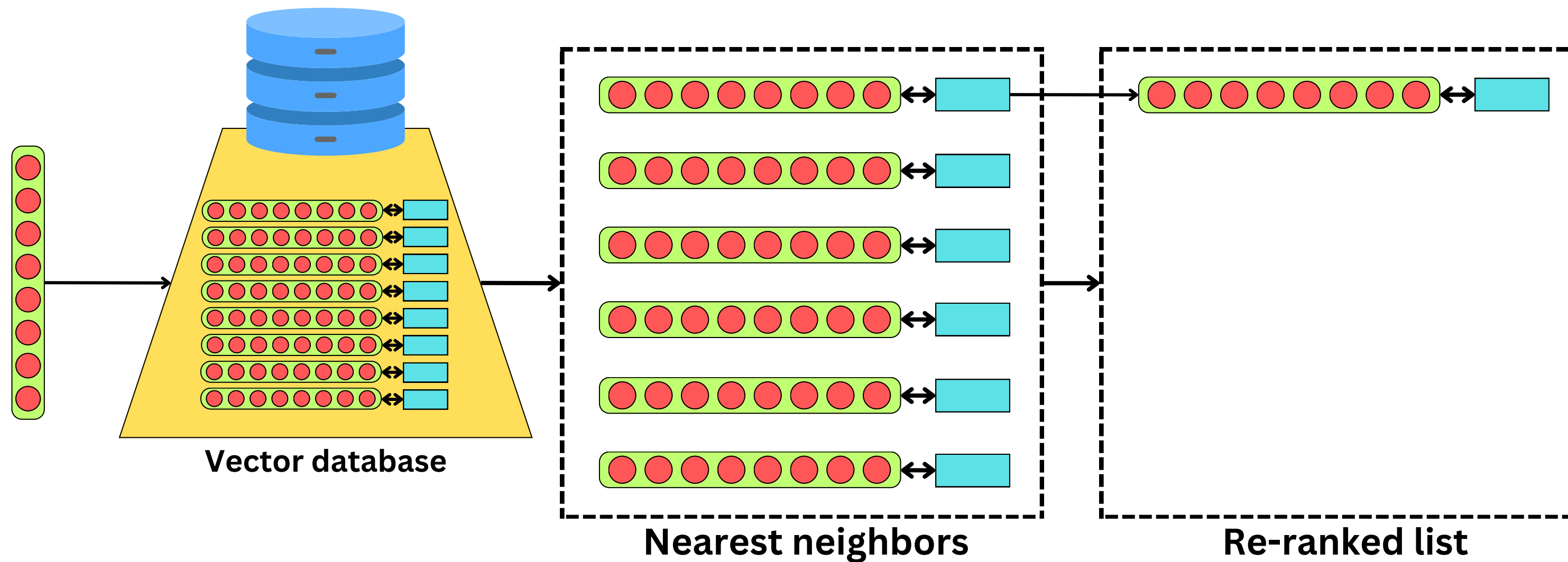
Maximal Marginal Relevance





Document selection optimization

Maximal Marginal Relevance

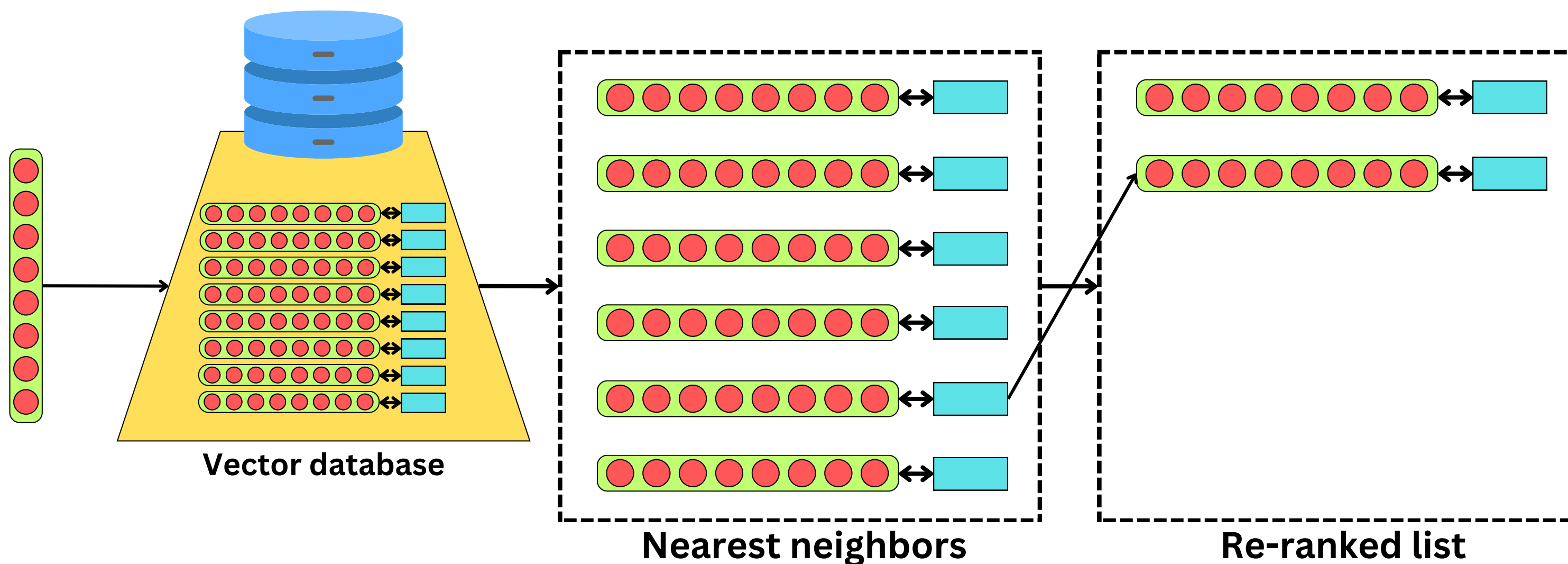




Document selection optimization

Maximal Marginal Relevance

$$MMR = \arg \max_{D_i \in R \setminus S} \left[\lambda \text{sim}(D_i, Q) - (1 - \lambda) \max_{D_j \in S} \text{sim}(D_i, D_j) \right]$$



The next vector needs to balance high similarity to the question
but low similarity to the vectors already in the list

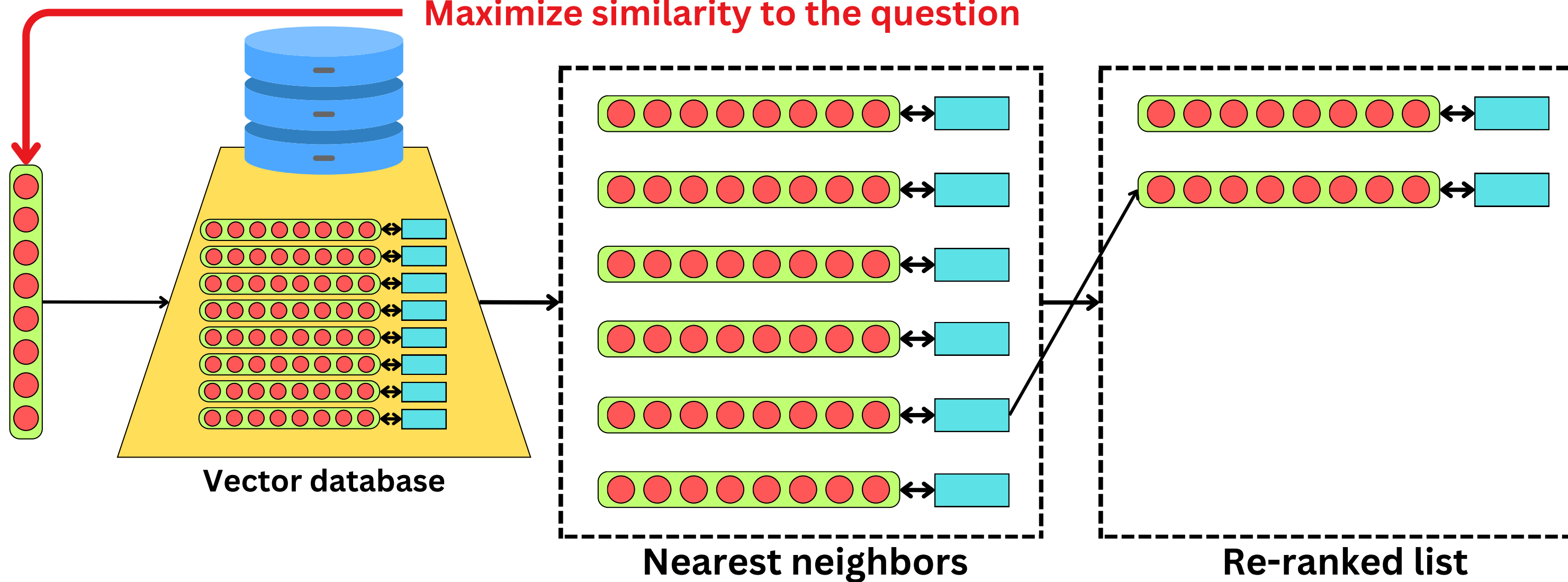


Document selection optimization

Maximal Marginal Relevance

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Maximize similarity to the question



The next vector needs to balance high similarity to the question
but low similarity to the vectors already in the list



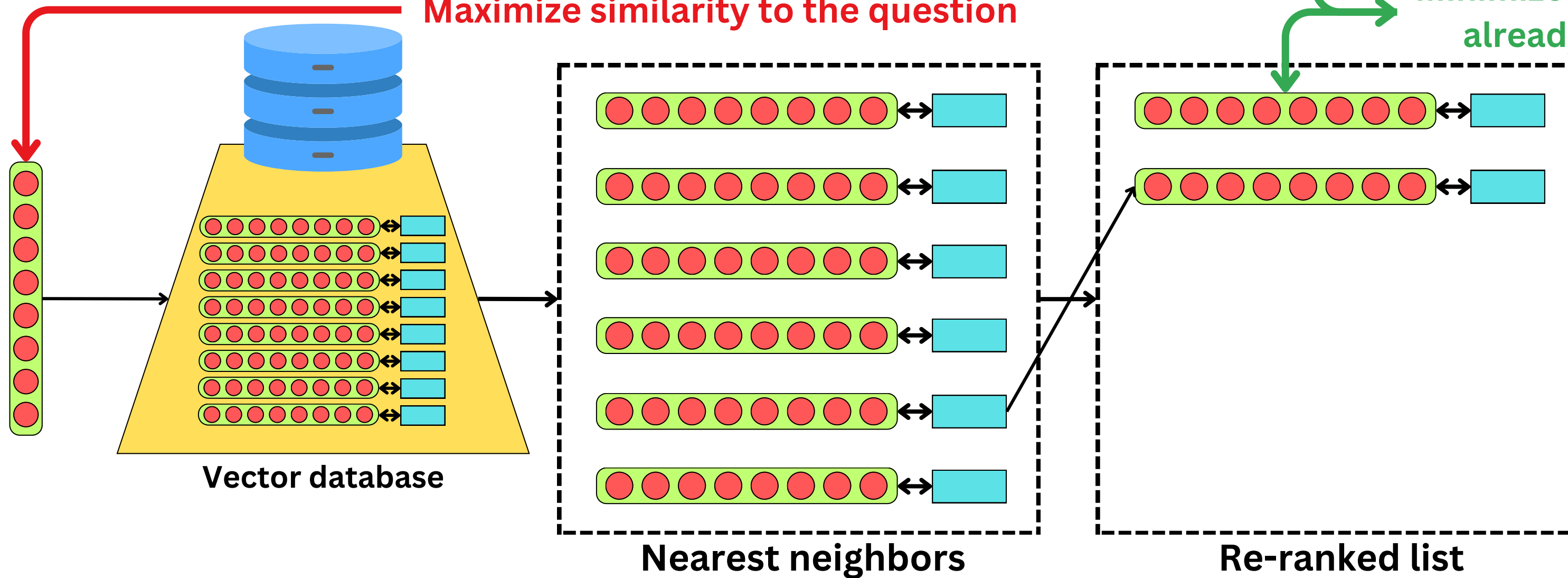
Document selection optimization

Maximal Marginal Relevance

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Maximize similarity to the question

Minimize similarity to the vectors already in the re-ranked list



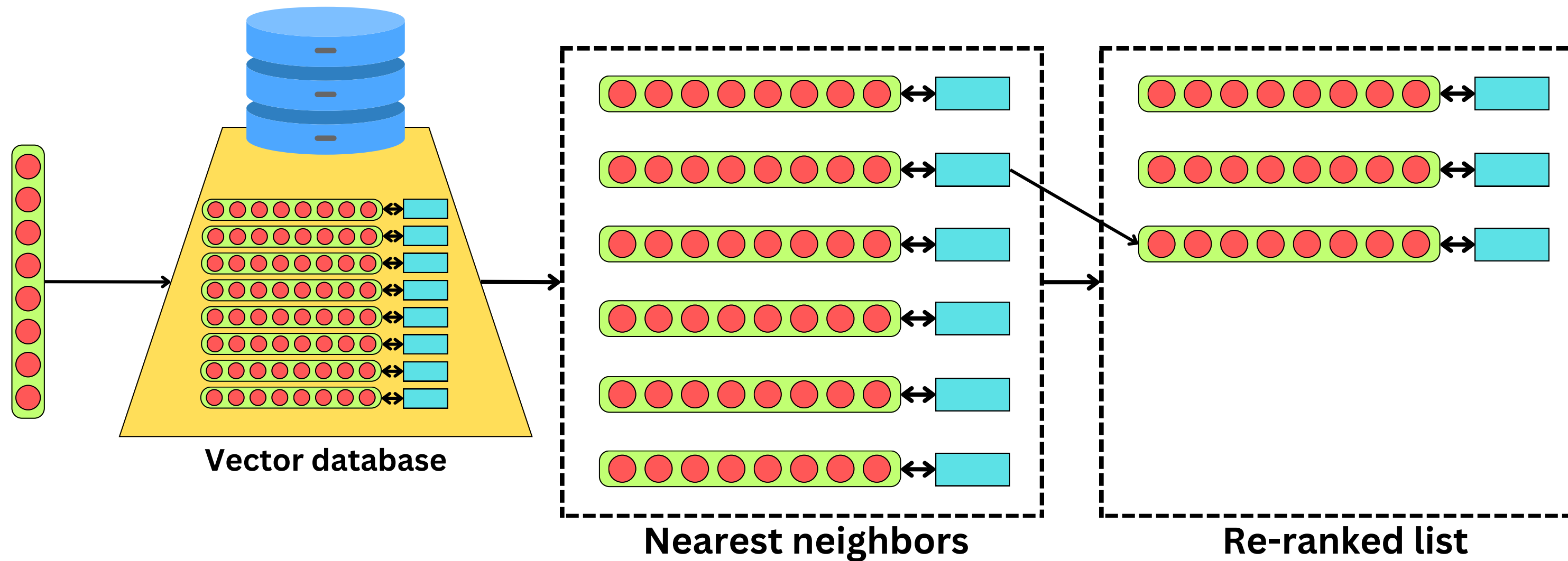
The next vector needs to balance high similarity to the question
but low similarity to the vectors already in the list



Document selection optimization

Maximal Marginal Relevance

$$MMR = \arg \max_{D_i \in R \setminus S} \left[\lambda \text{sim}(D_i, Q) - (1 - \lambda) \max_{D_j \in S} \text{sim}(D_i, D_j) \right]$$



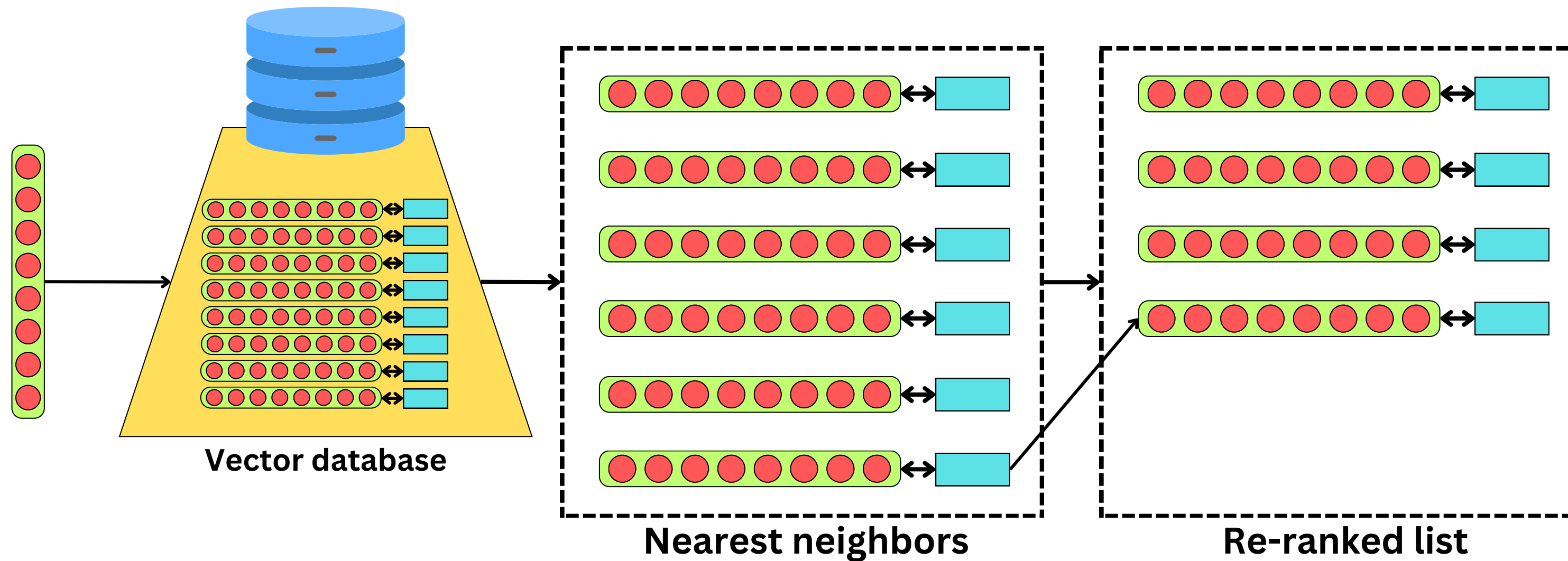
The next vector needs to balance high similarity to the question
but low similarity to the vectors already in the list



Document selection optimization

Maximal Marginal Relevance

$$MMR = \arg \max_{D_i \in R \setminus S} \left[\lambda \text{sim}(D_i, Q) - (1 - \lambda) \max_{D_j \in S} \text{sim}(D_i, D_j) \right]$$



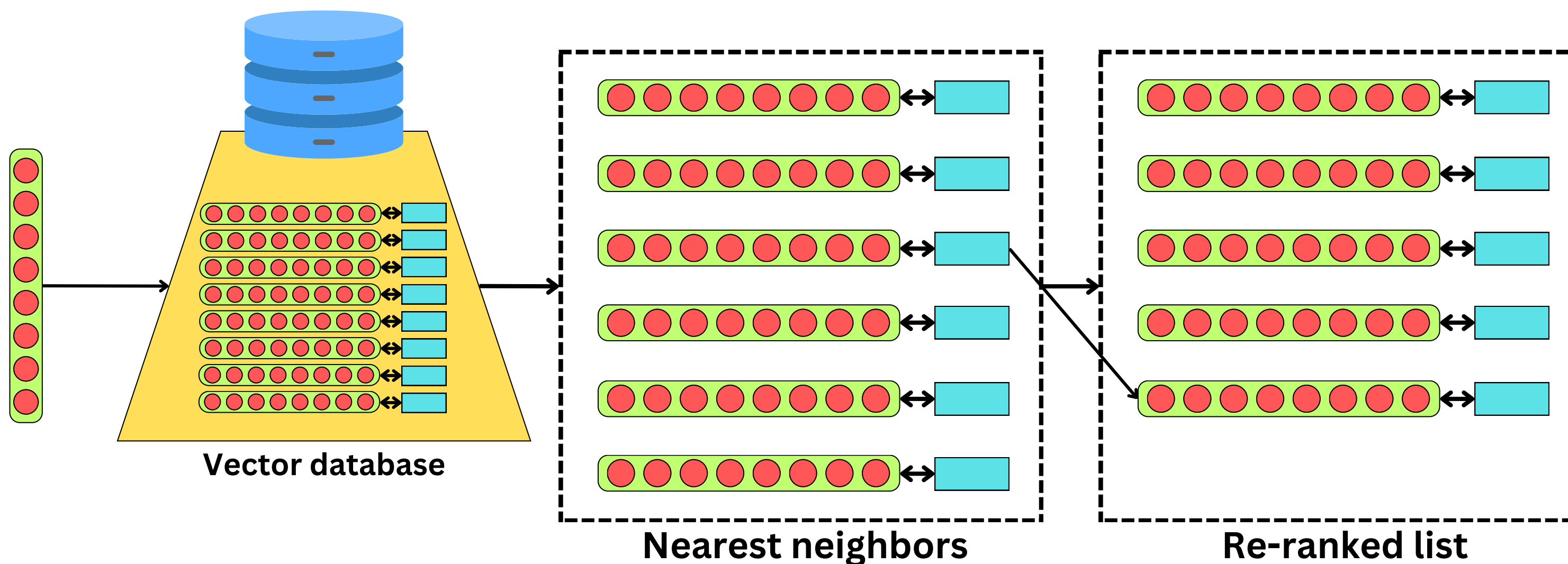
The next vector needs to balance high similarity to the question
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Document selection optimization

Maximal Marginal Relevance

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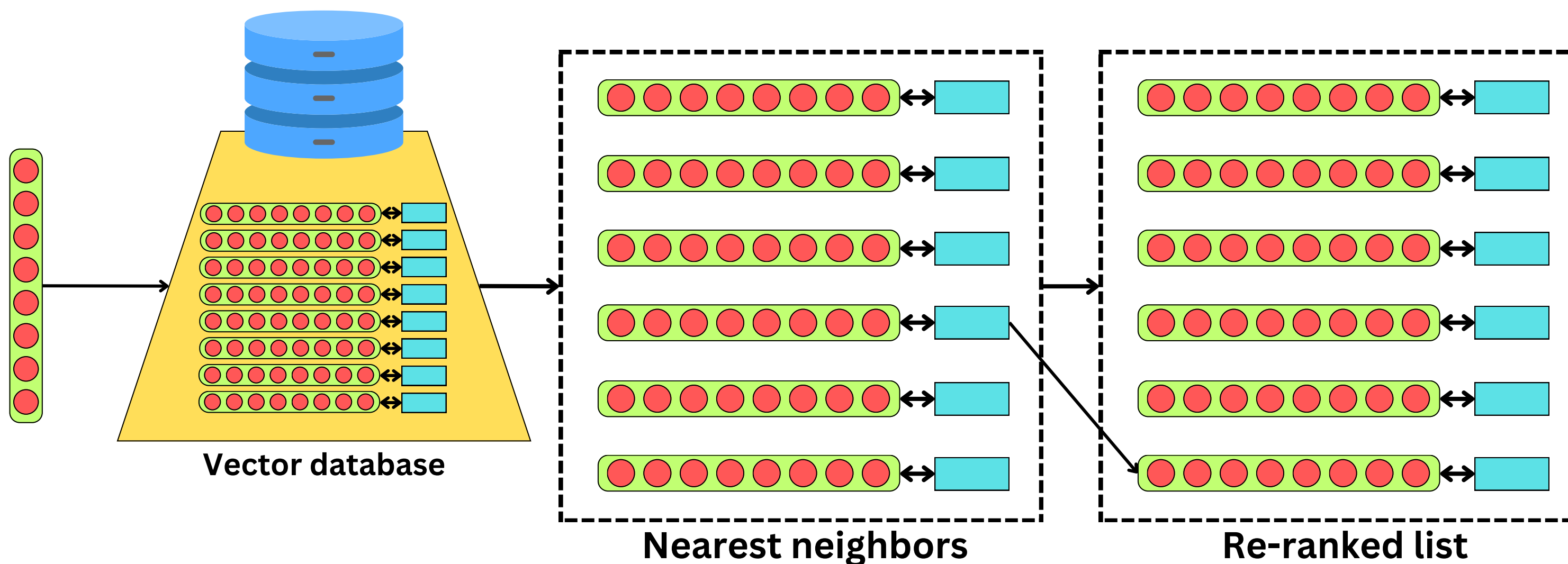
The next vector needs to balance high similarity to the question
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Document selection optimization

Maximal Marginal Relevance

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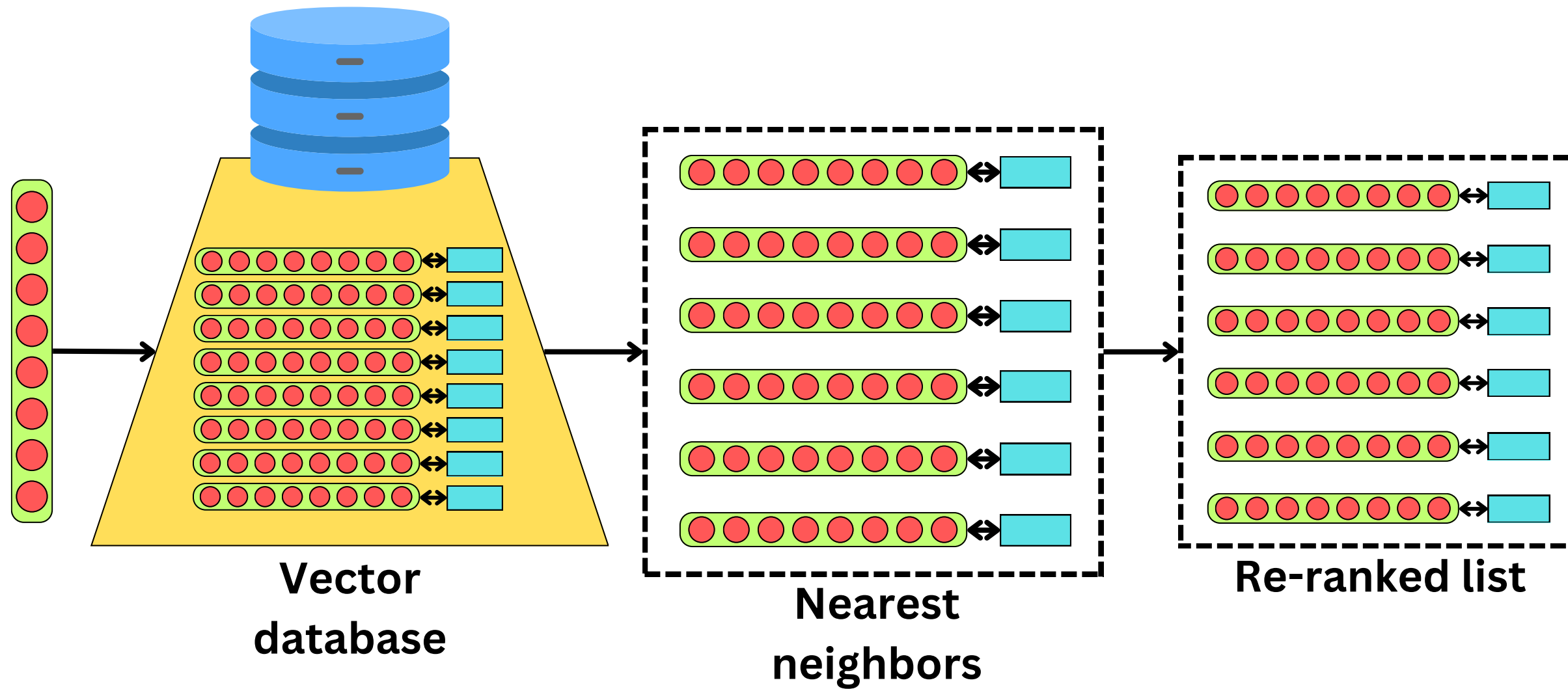


The next vector needs to balance high similarity to the question
but low similarity to the vectors already in the list



Document selection optimization

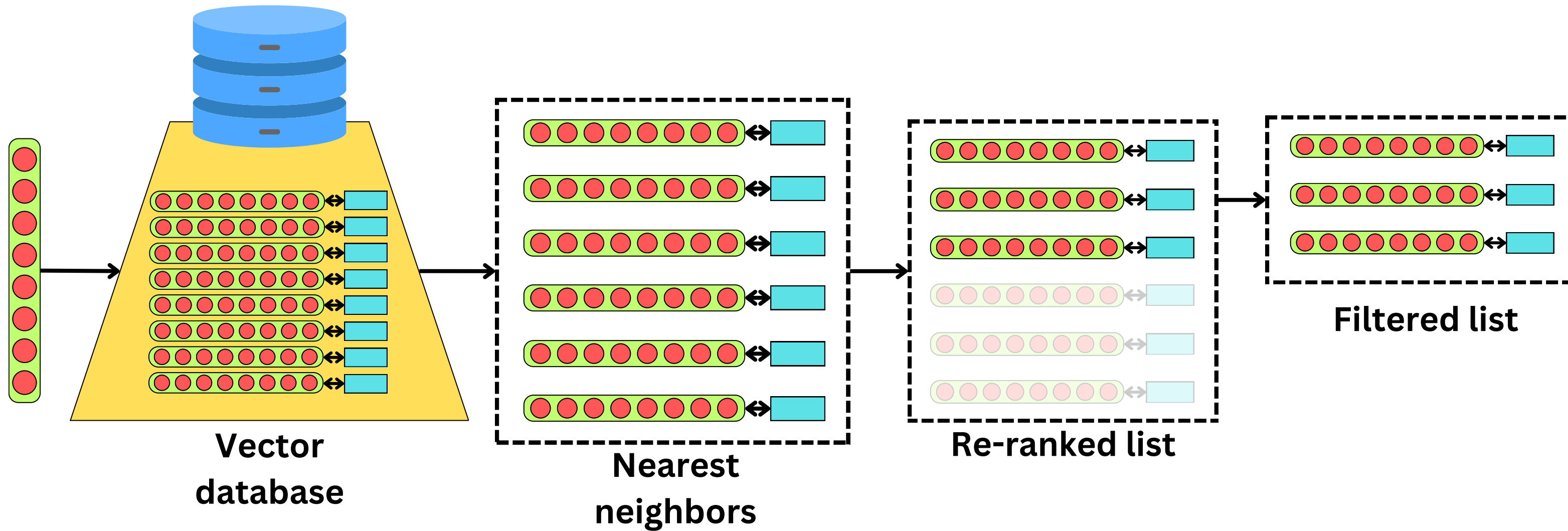
Maximal Marginal Relevance





Document selection optimization

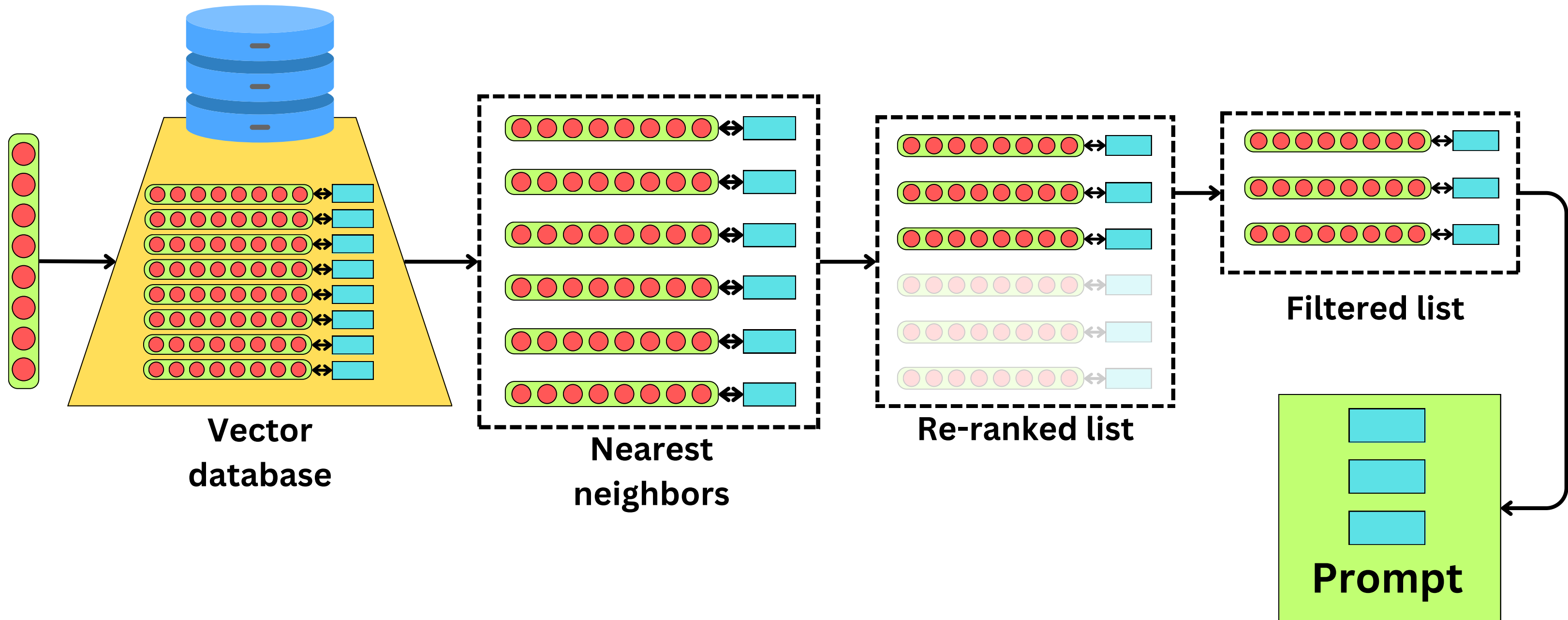
Maximal Marginal Relevance





Document selection optimization

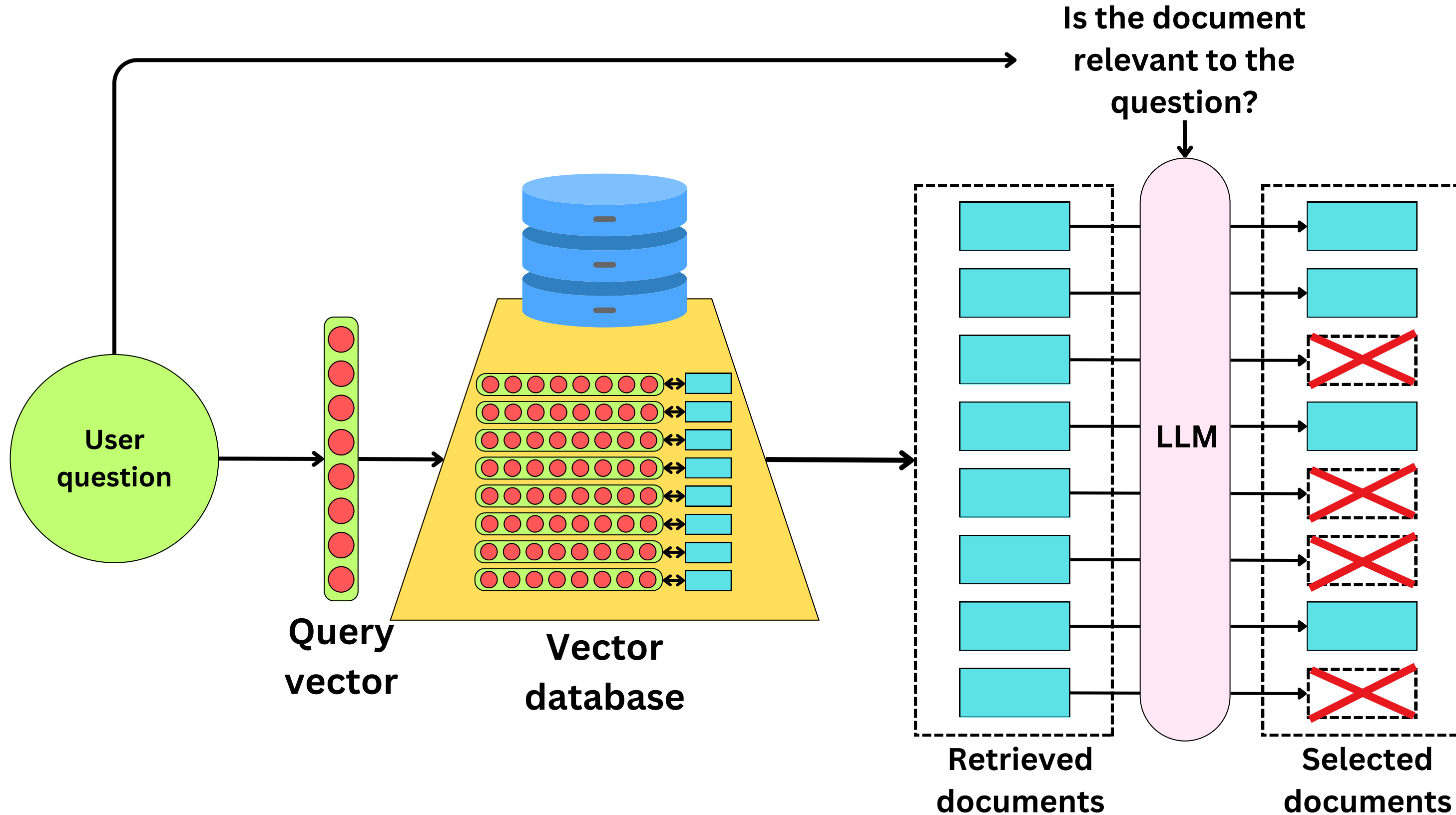
Maximal Marginal Relevance





Document selection optimization

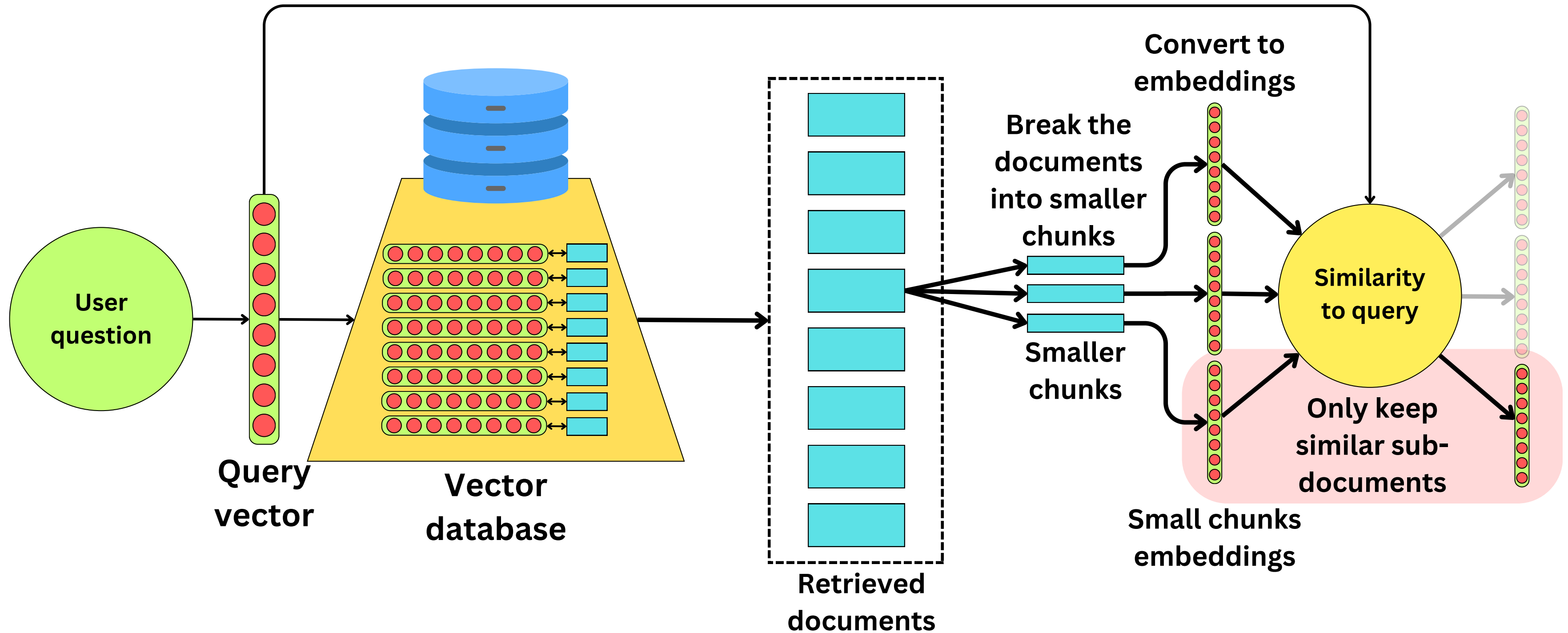
Contextual compression





Document selection optimization

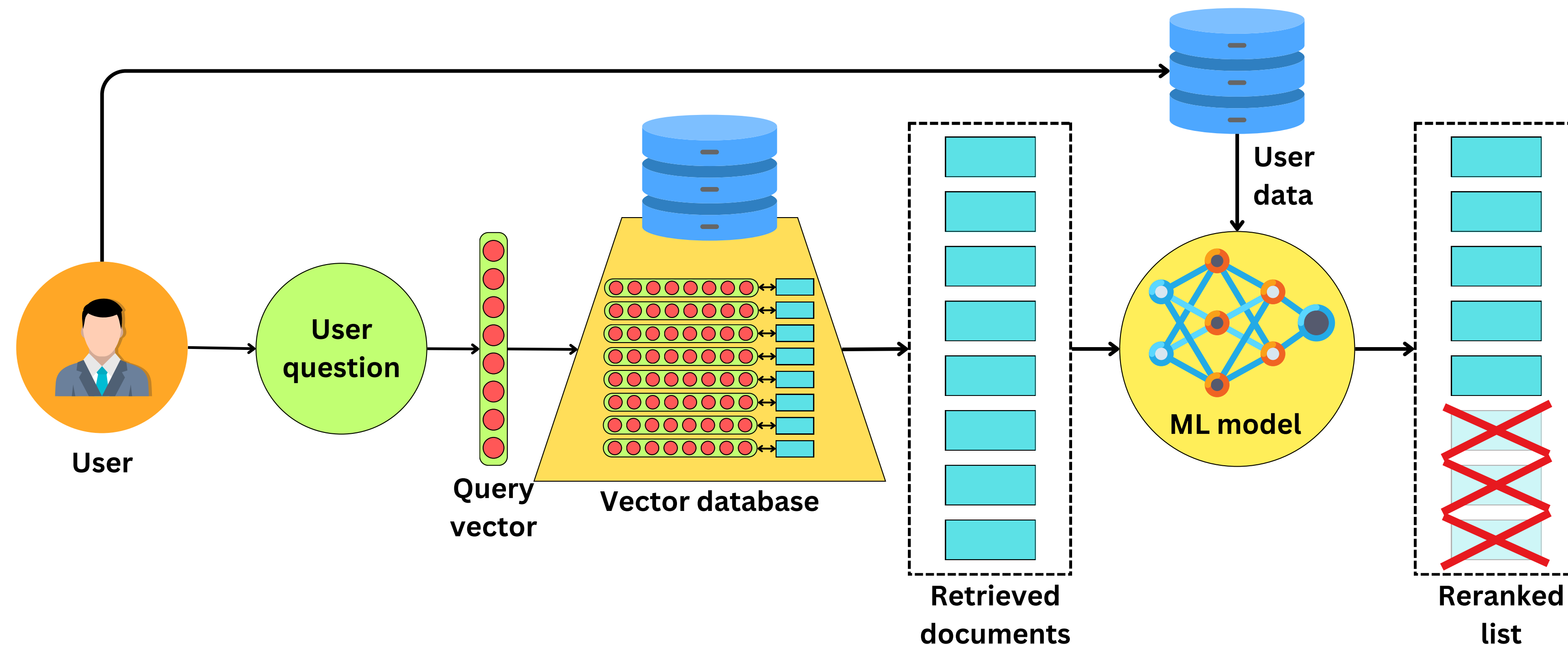
Contextual compression





Document selection optimization

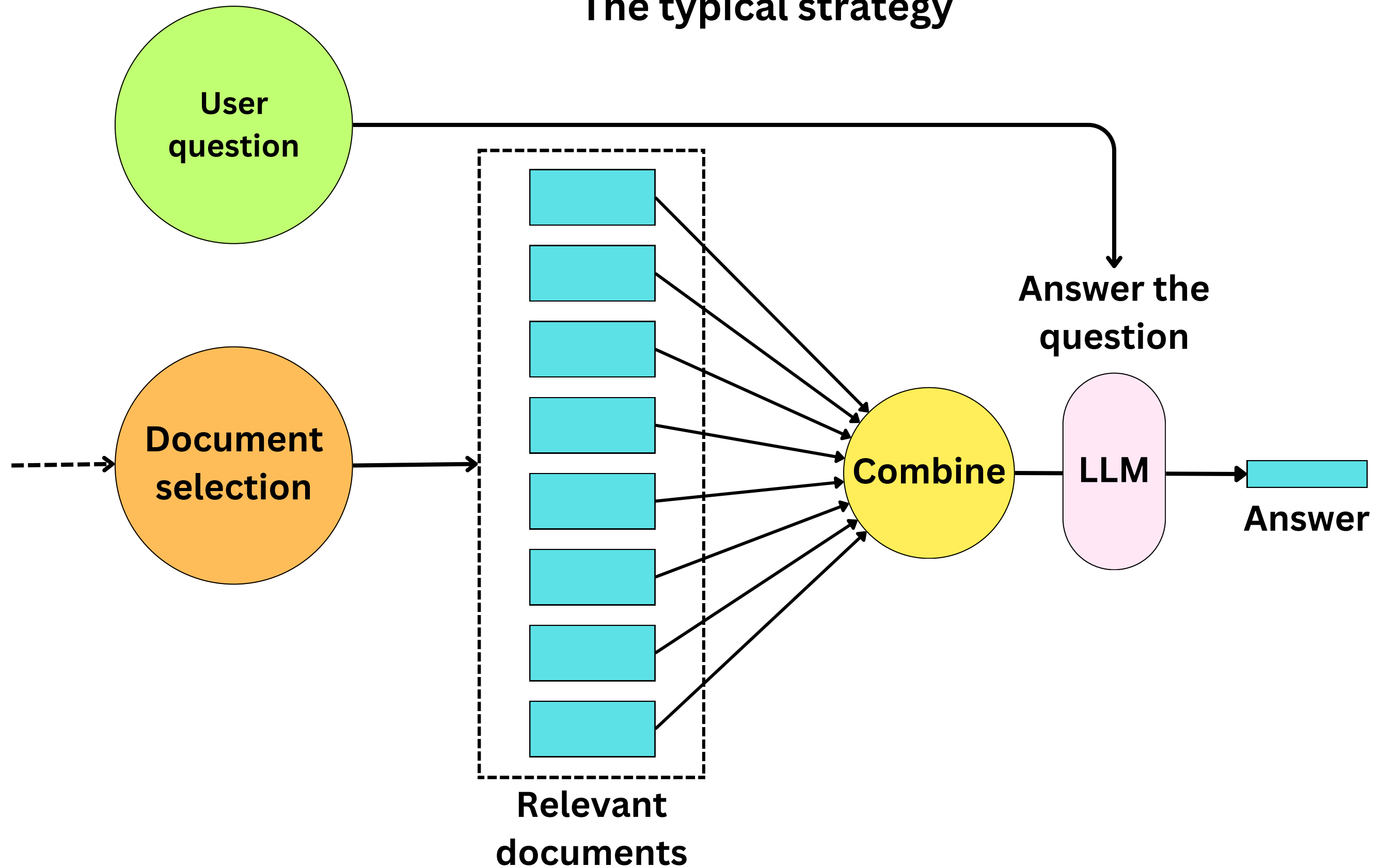
Personalization





Context Optimization

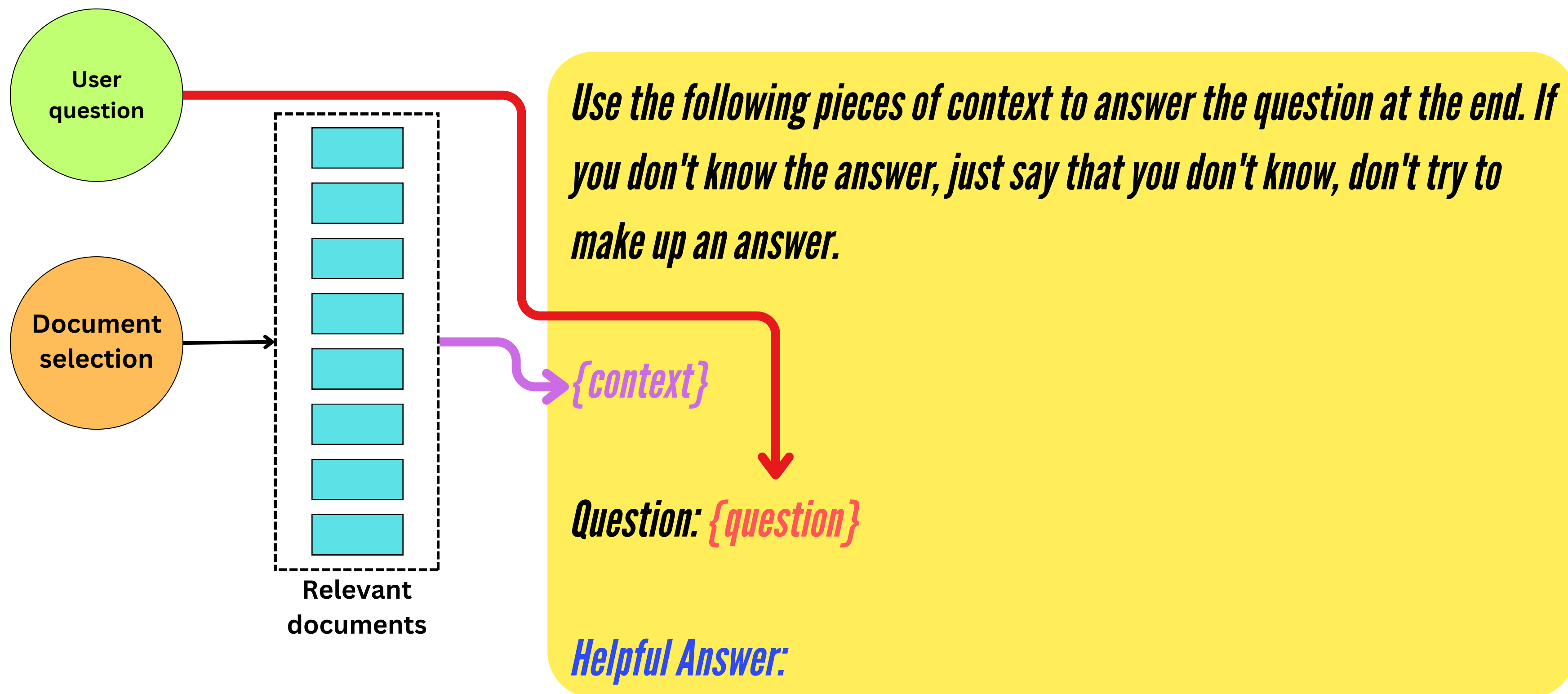
The typical strategy





Context Optimization

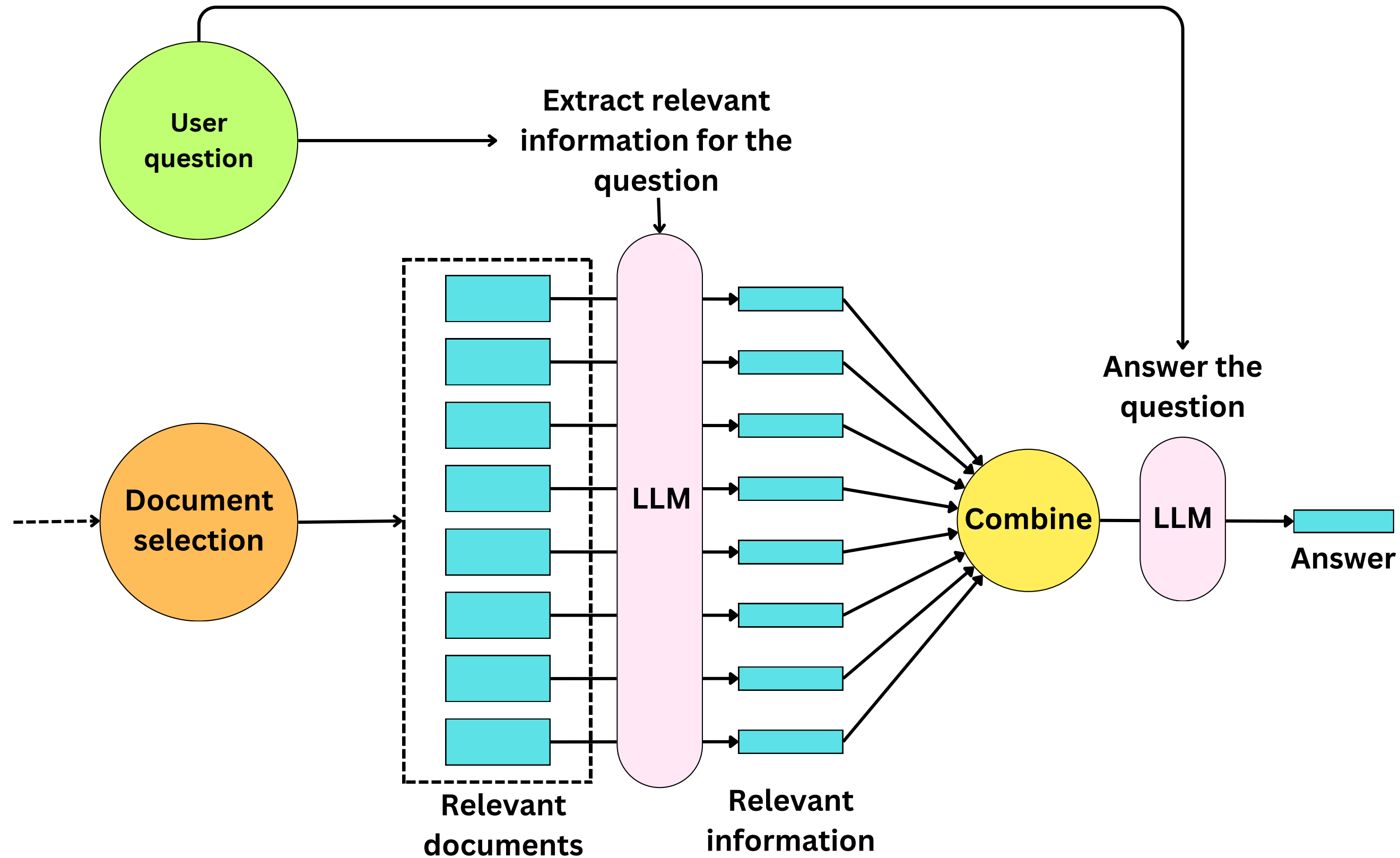
The typical strategy





Context Optimization

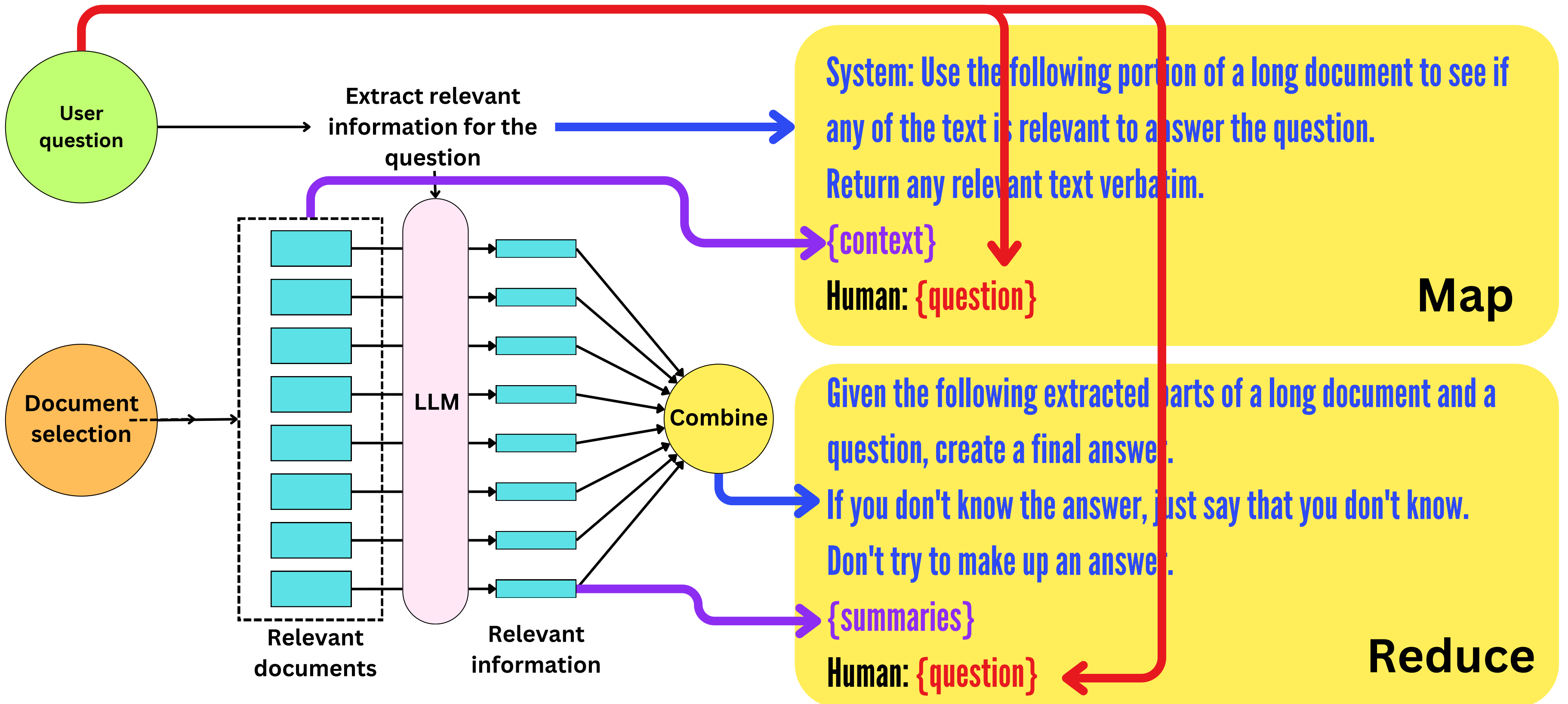
The Map-Reduce strategy





Context Optimization

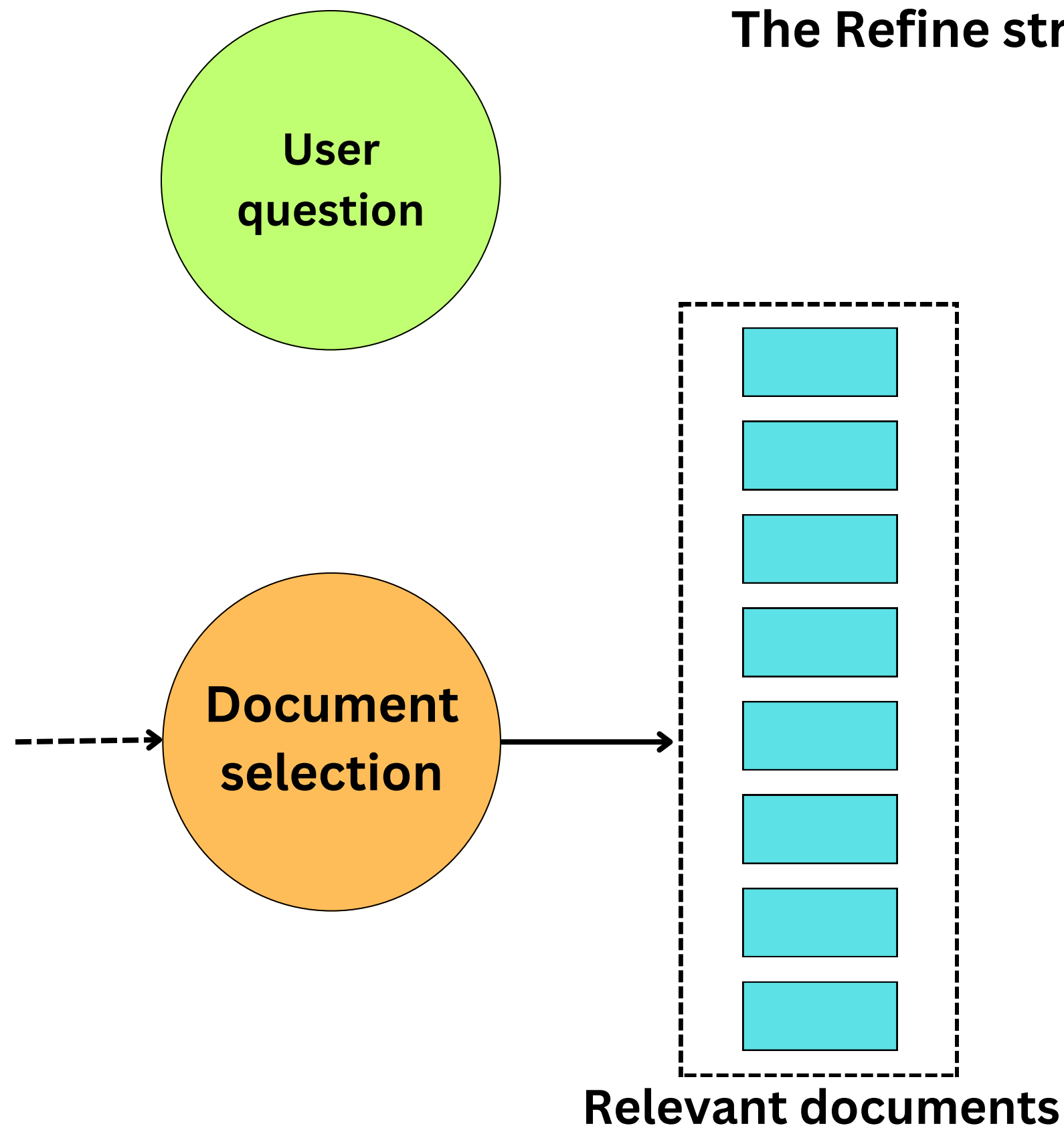
The Map-Reduce strategy





Context Optimization

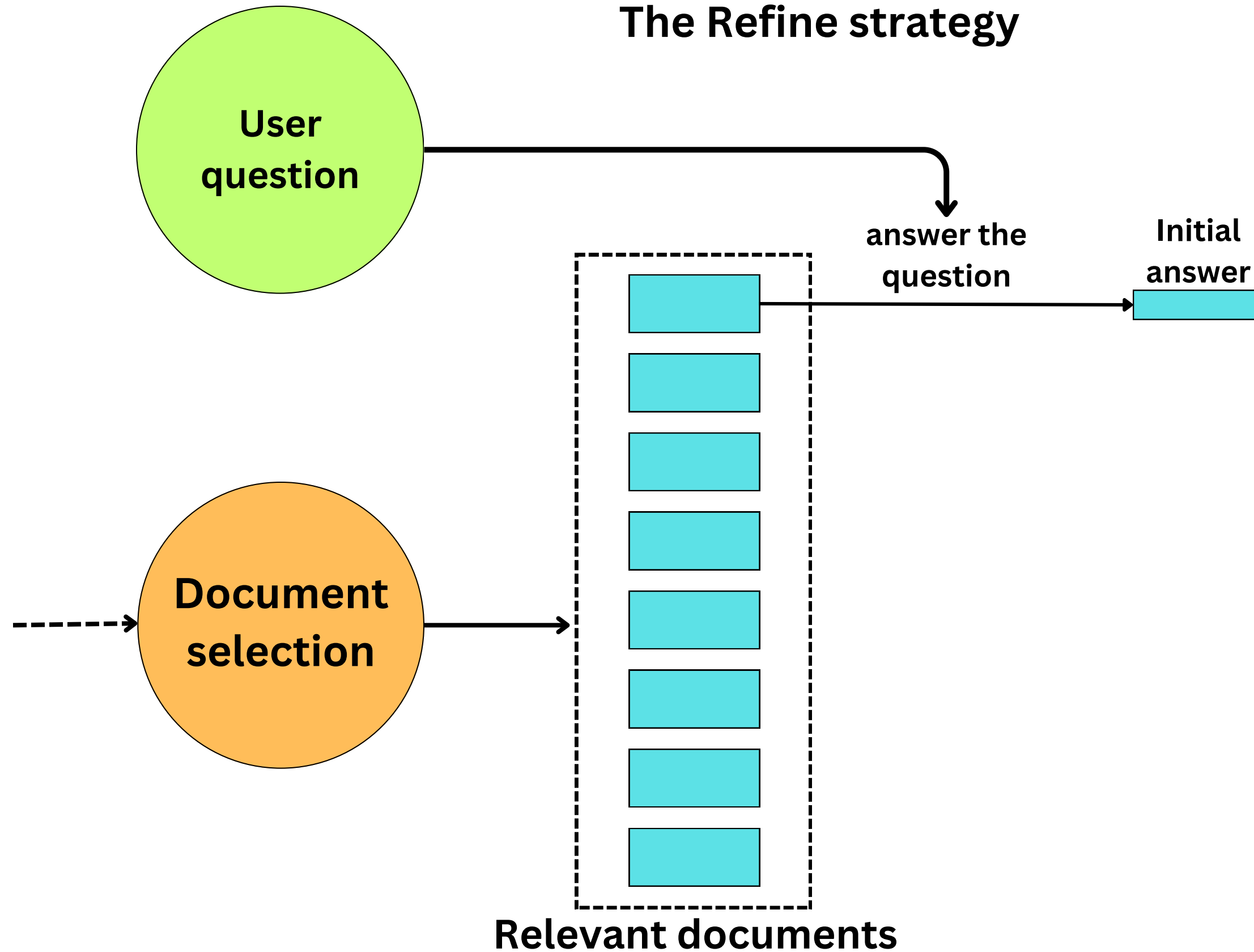
The Refine strategy





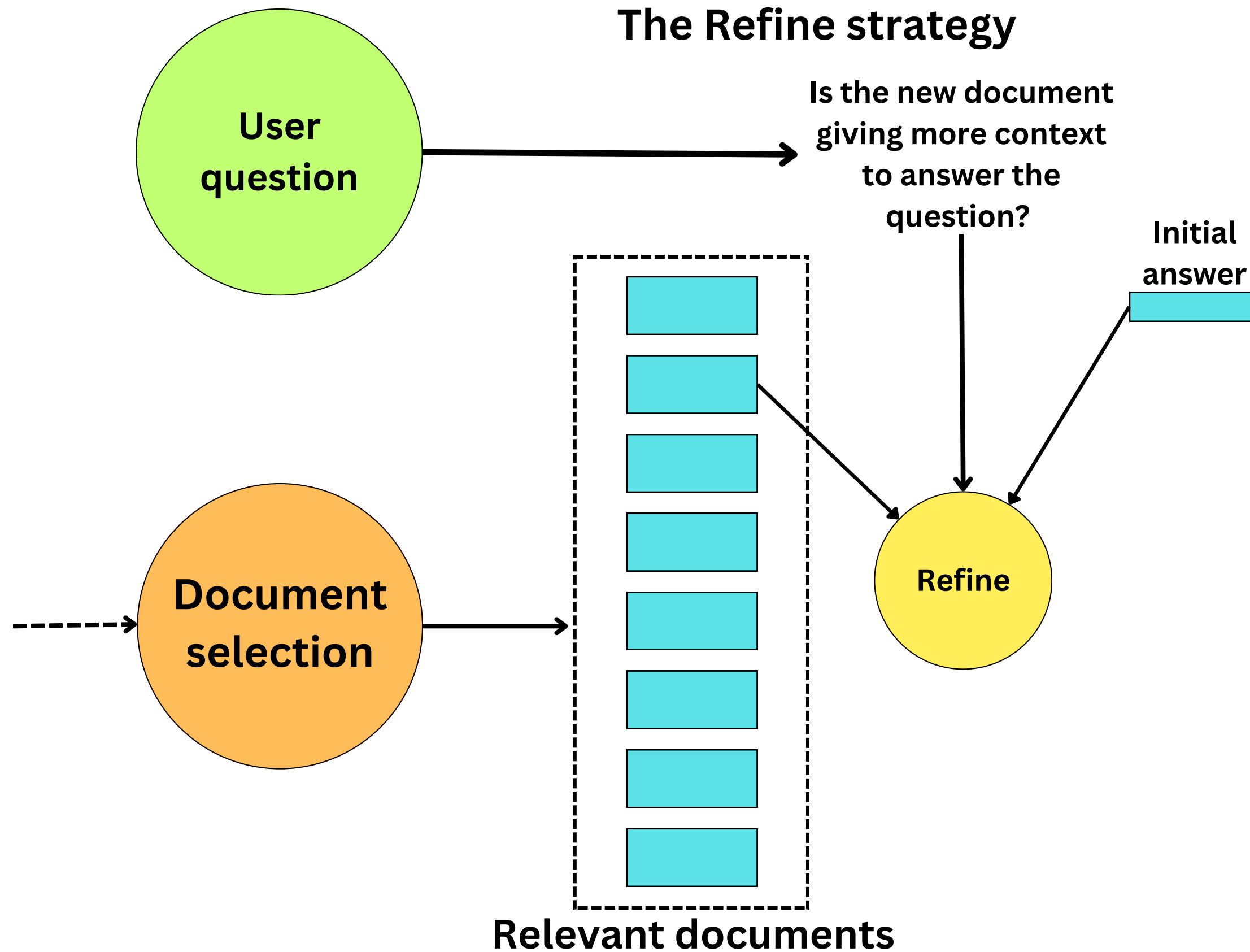
Context Optimization

The Refine strategy





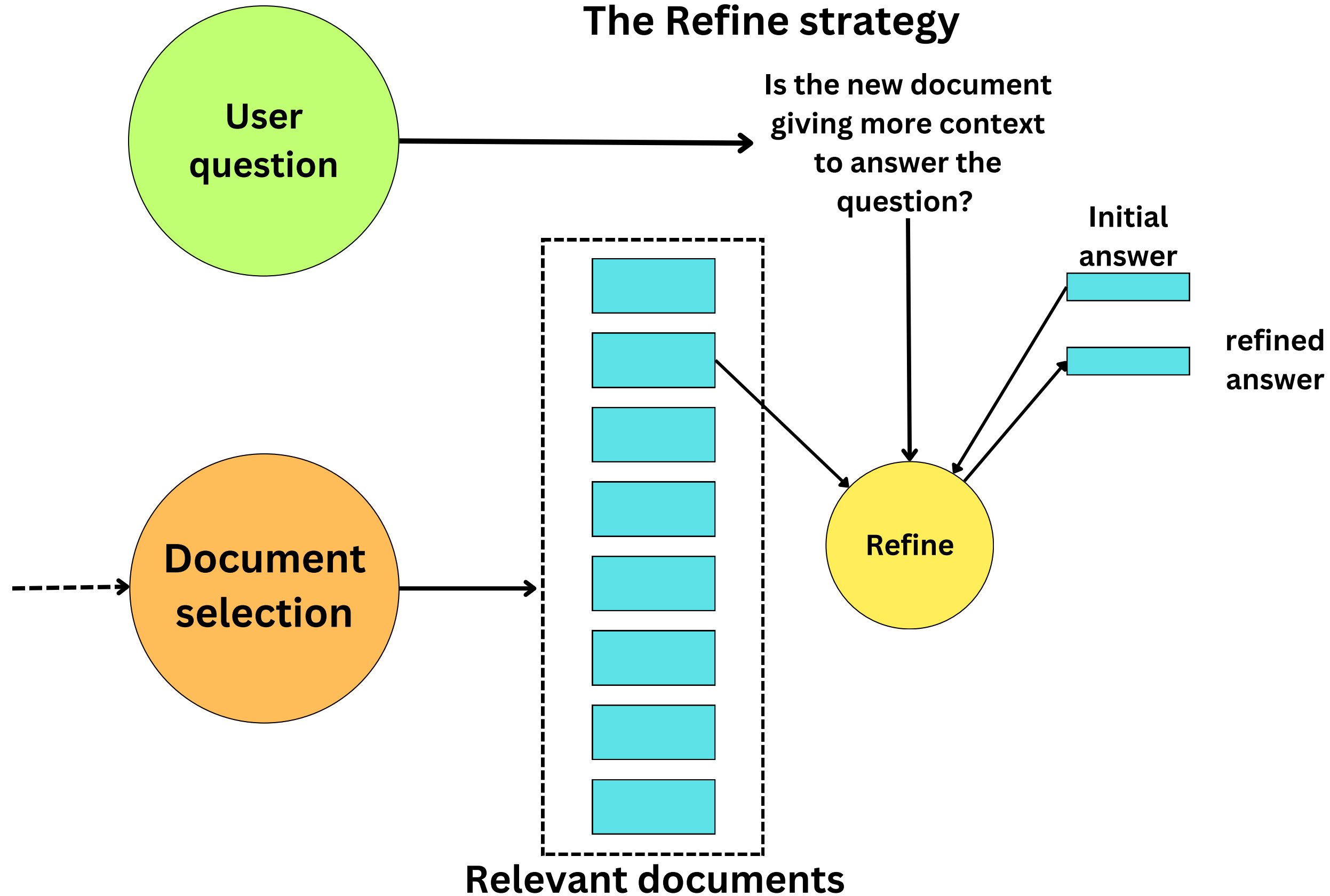
Context Optimization





Context Optimization

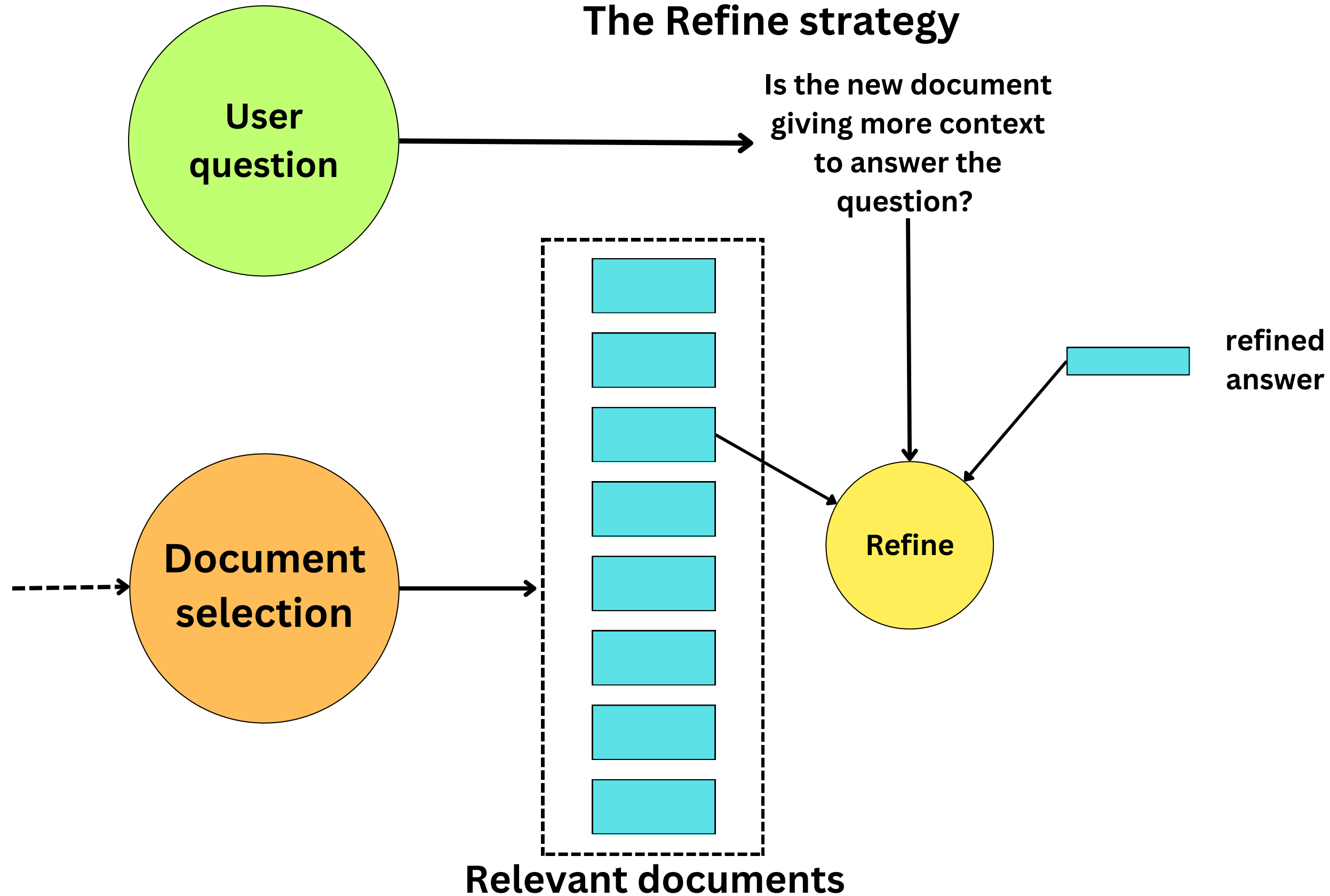
The Refine strategy





Context Optimization

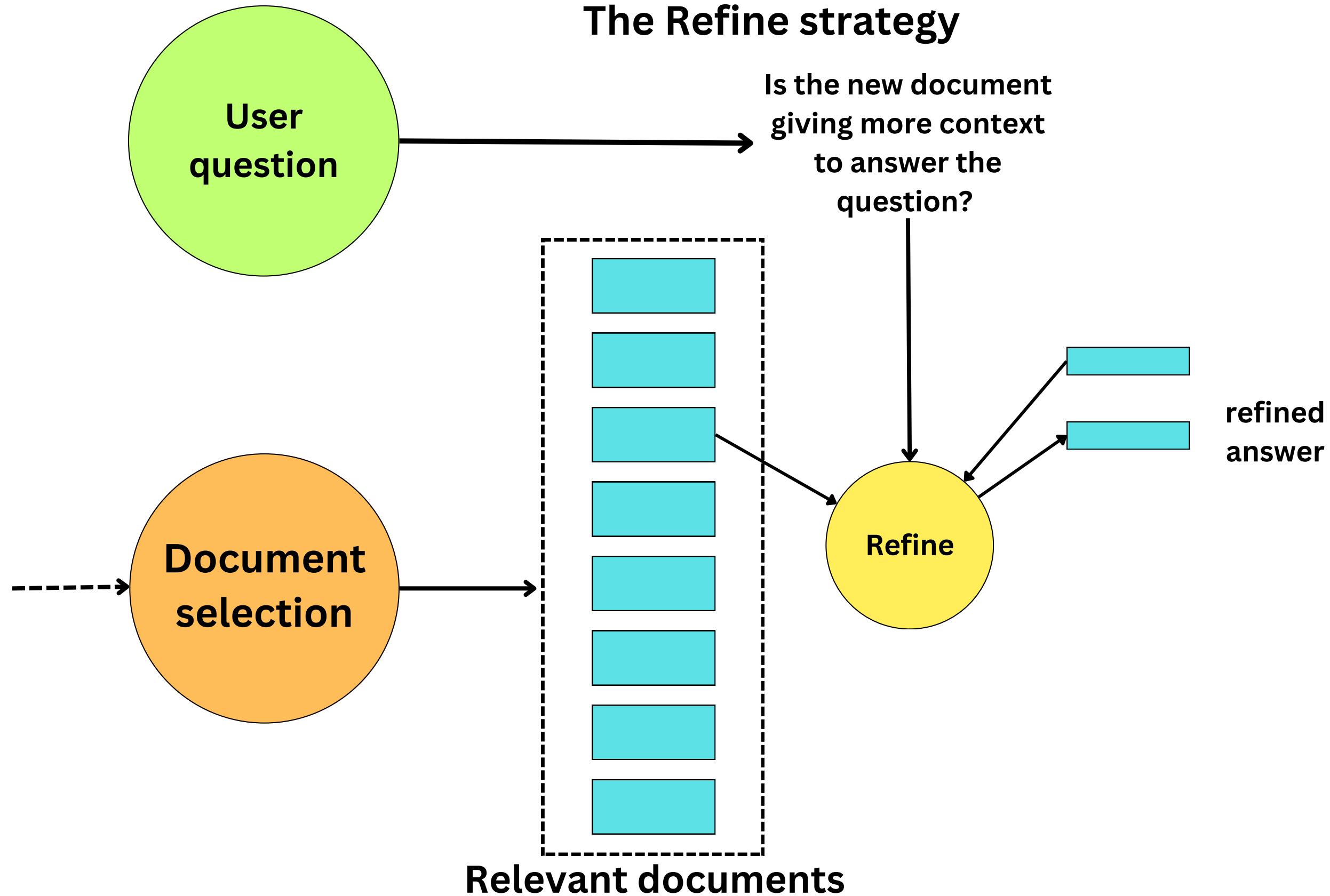
The Refine strategy





Context Optimization

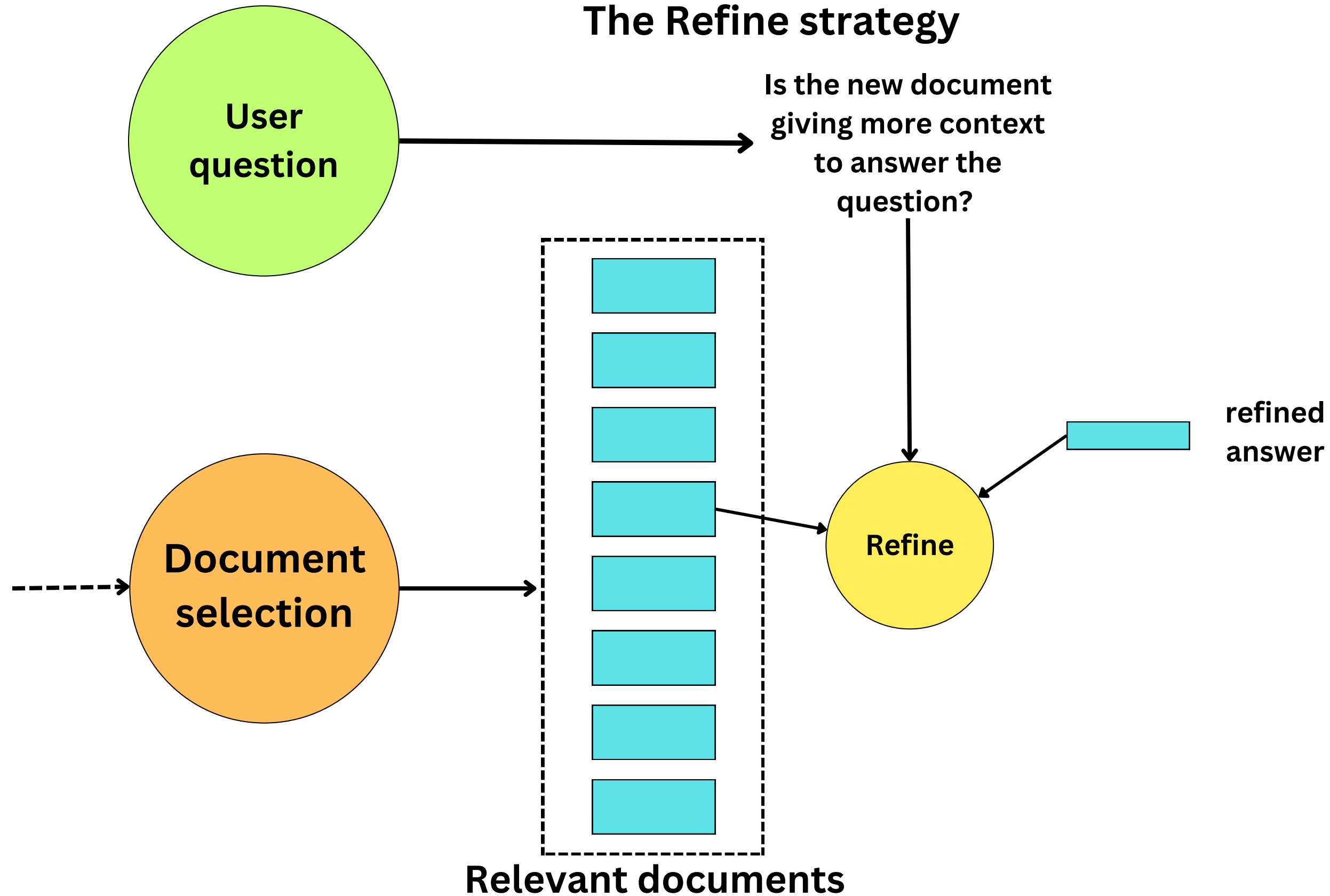
The Refine strategy





Context Optimization

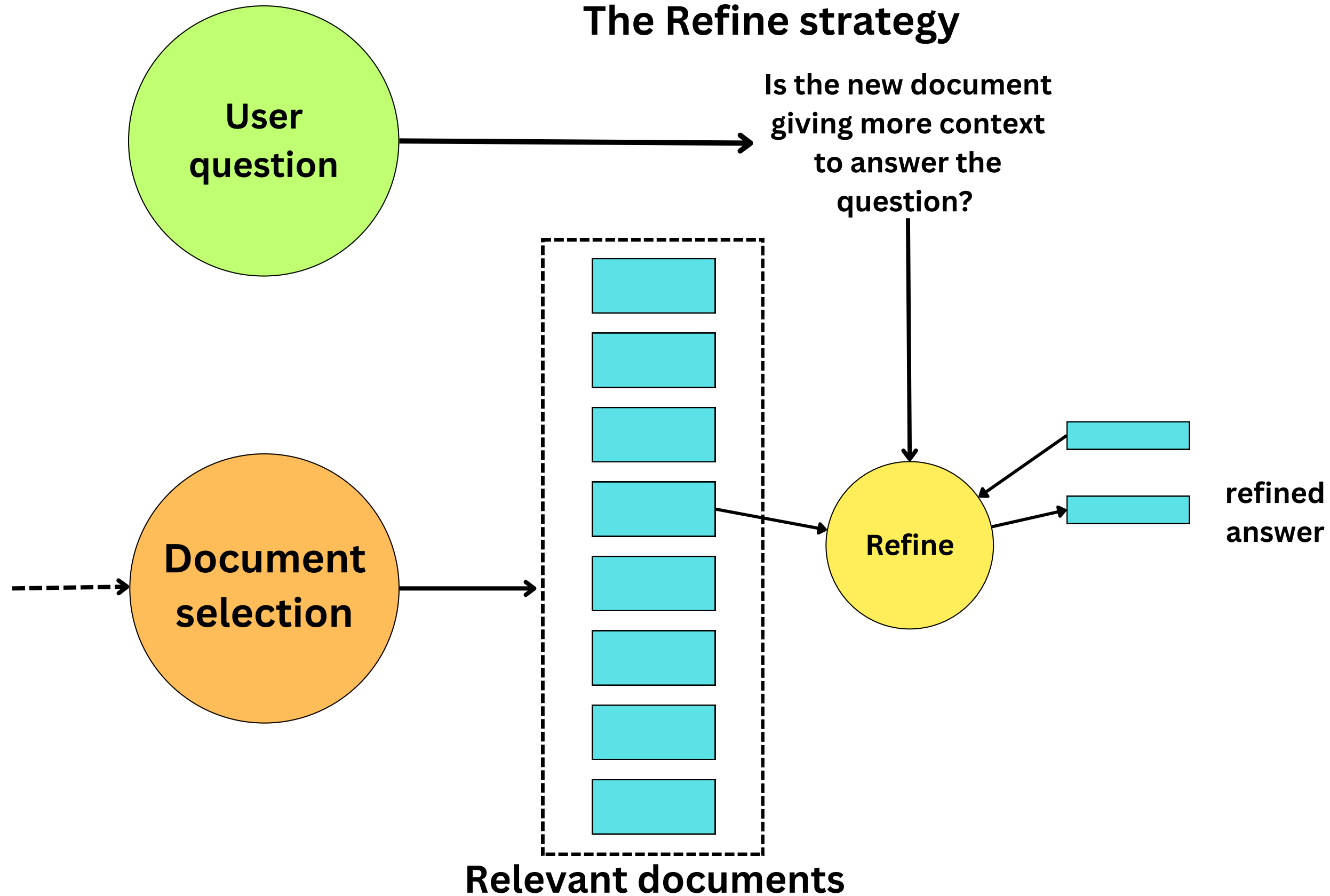
The Refine strategy





Context Optimization

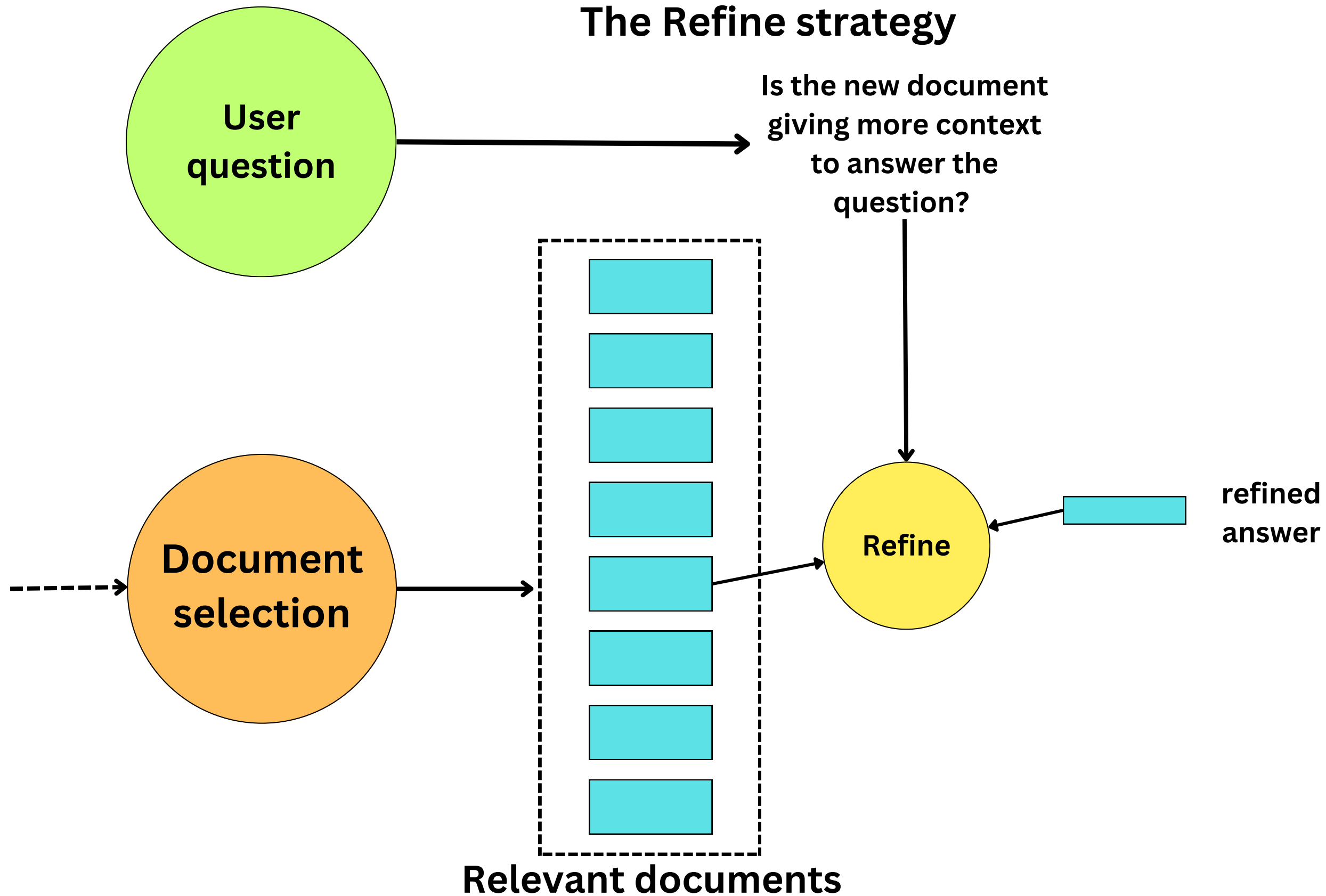
The Refine strategy





Context Optimization

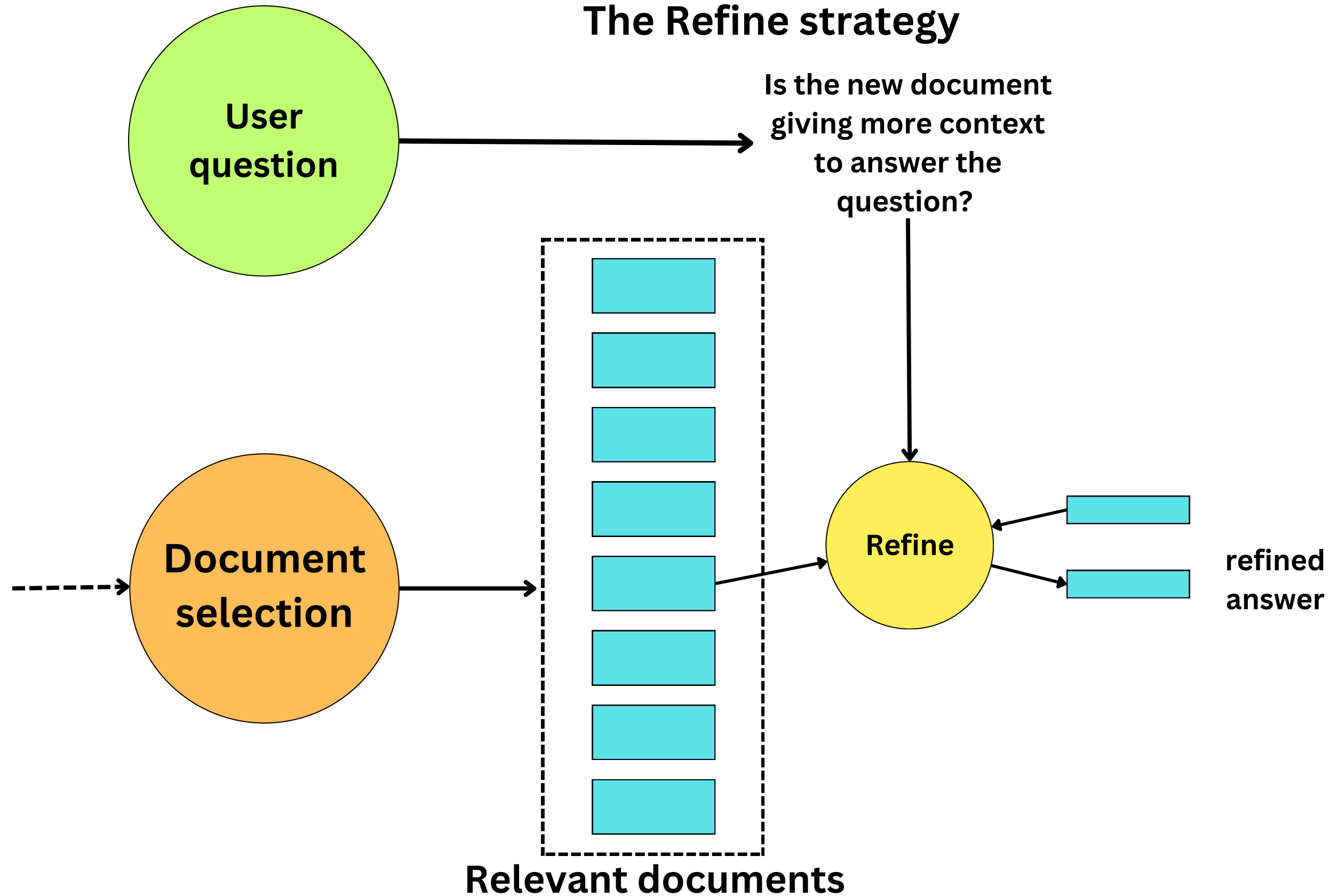
The Refine strategy





Context Optimization

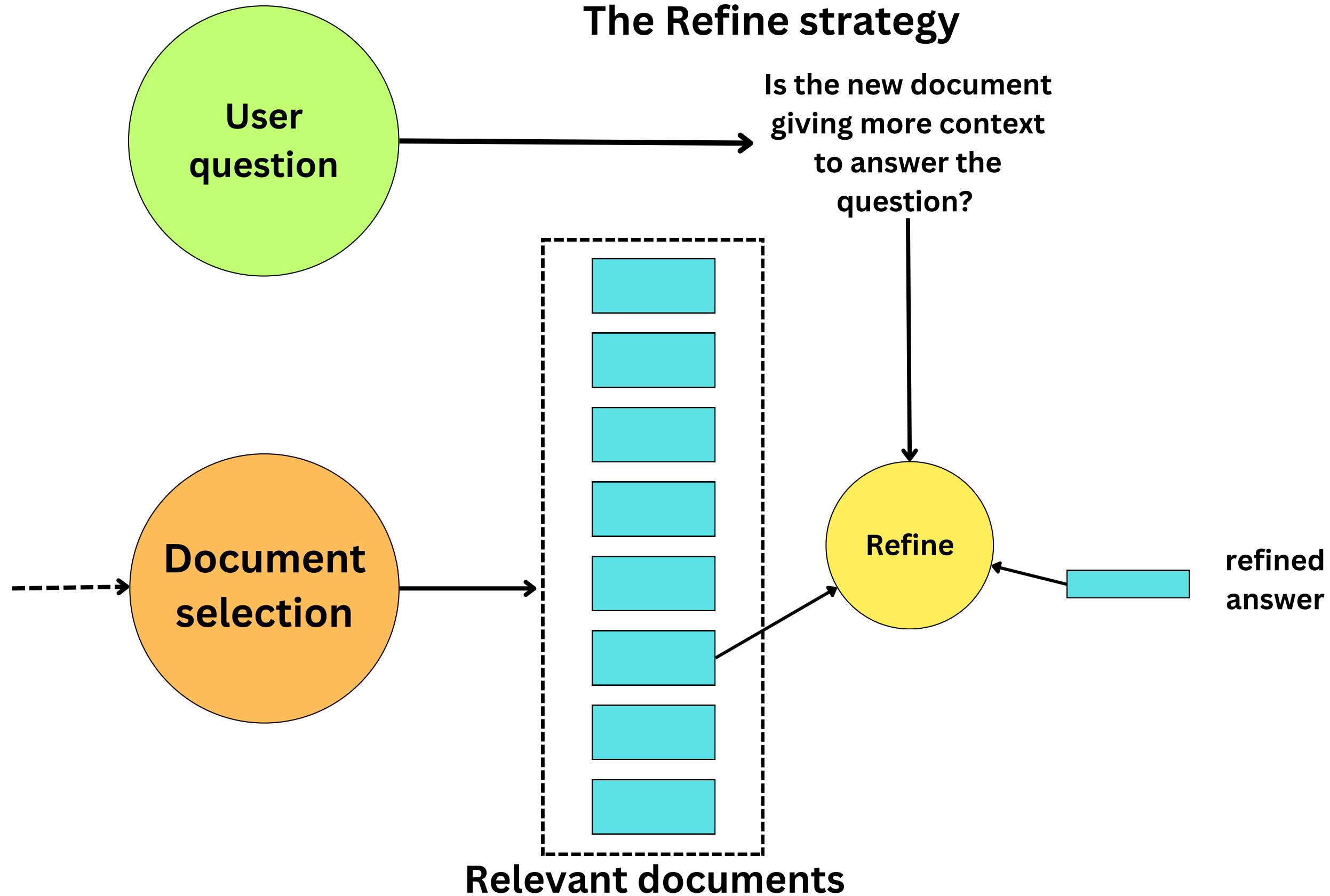
The Refine strategy





Context Optimization

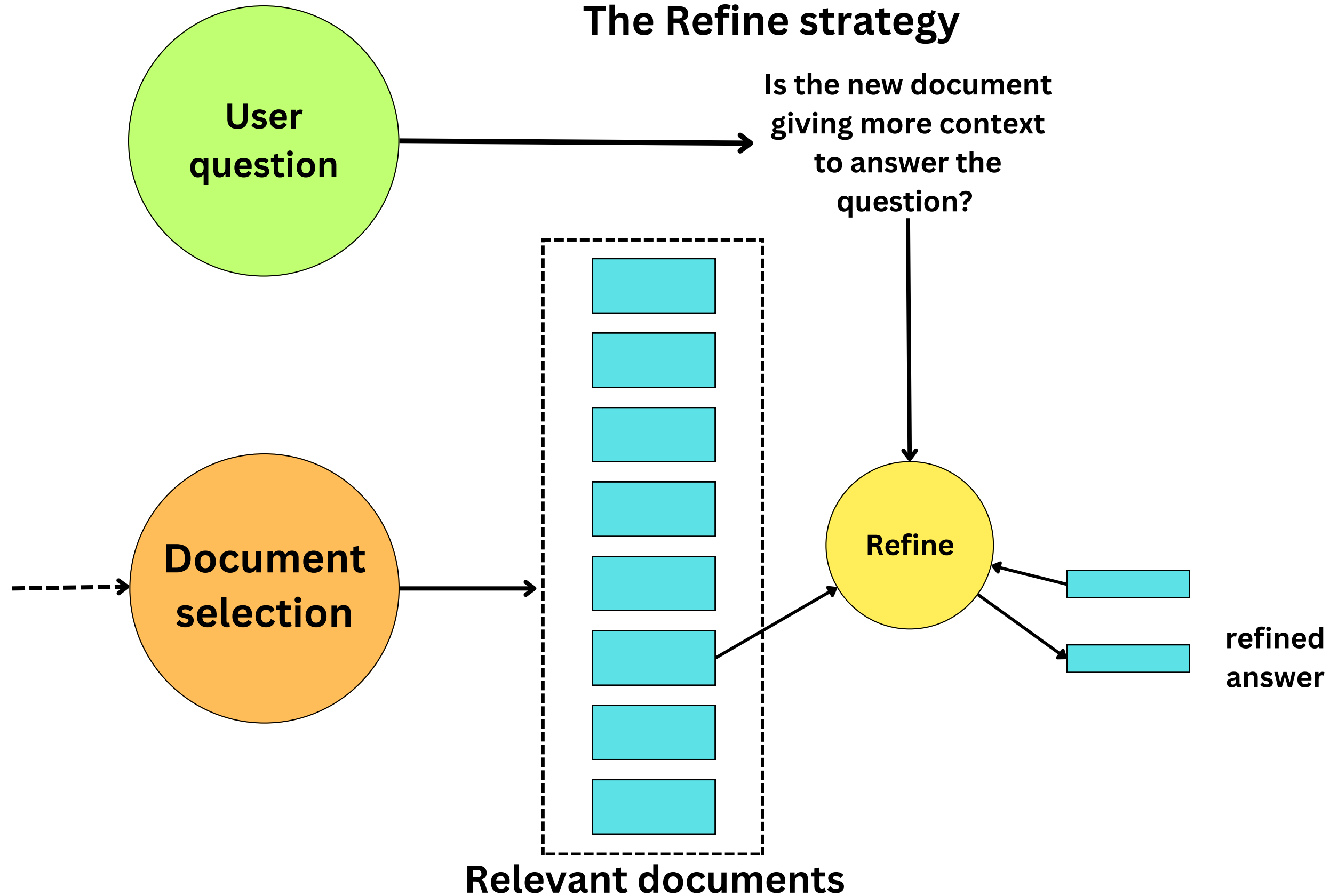
The Refine strategy





Context Optimization

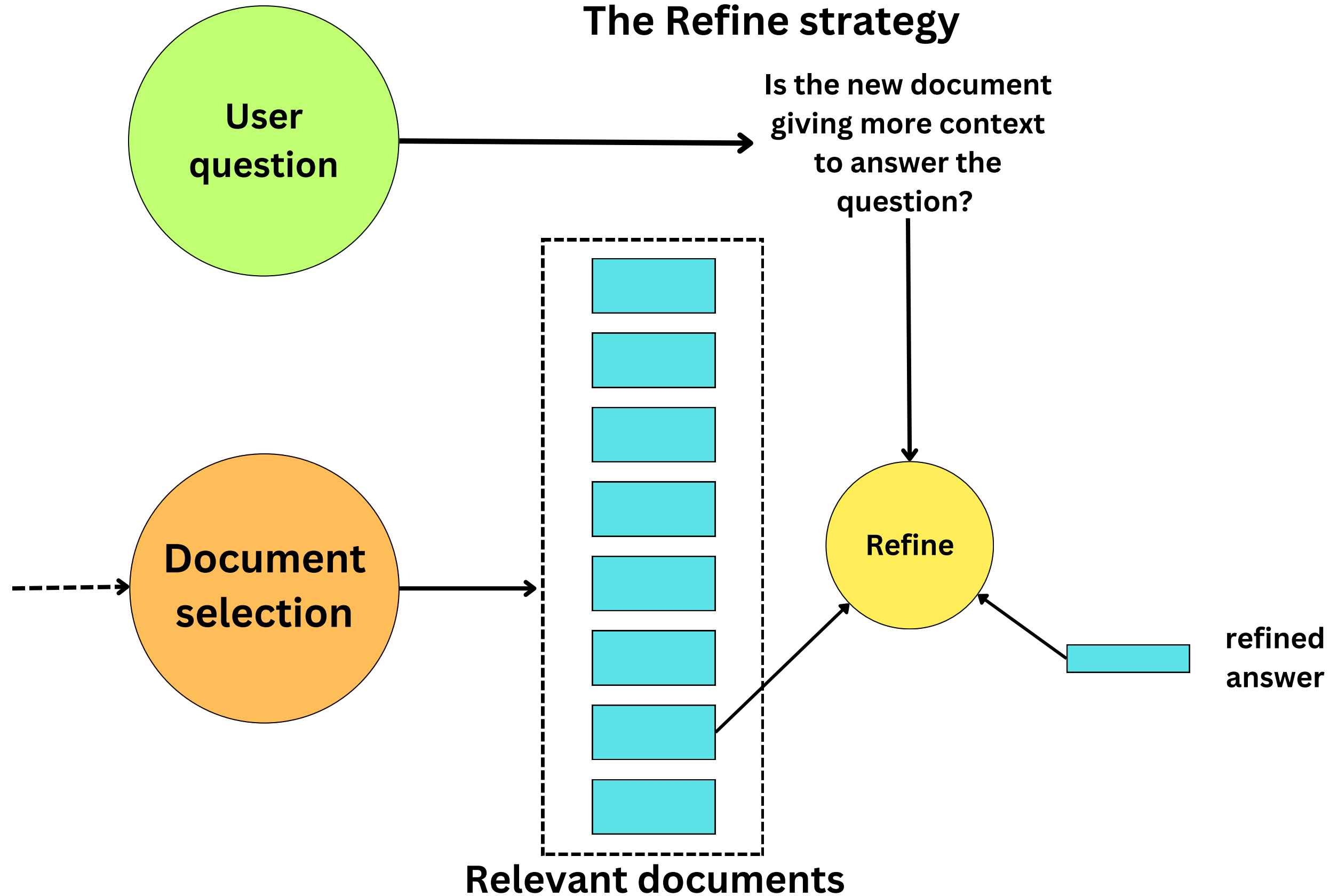
The Refine strategy





Context Optimization

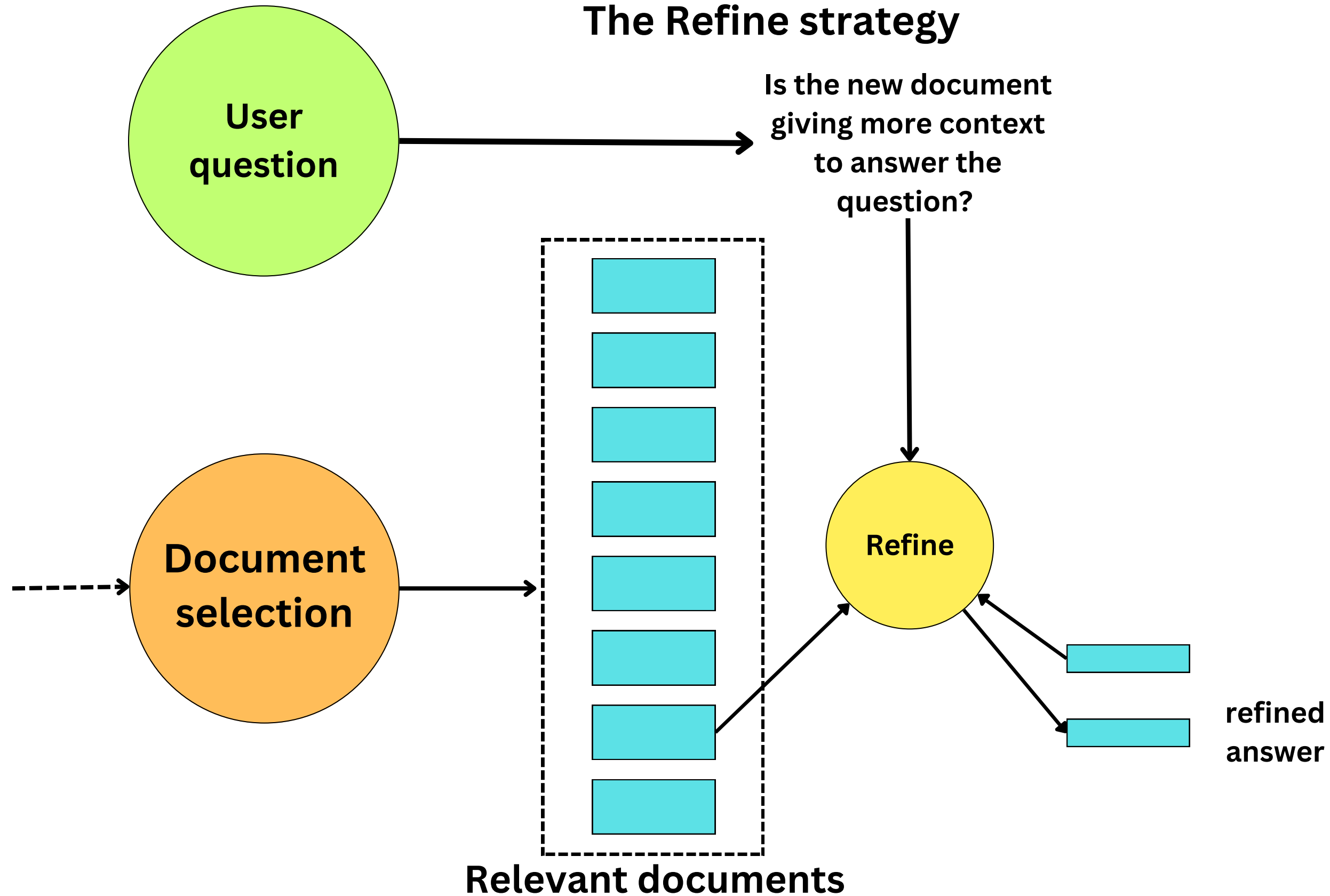
The Refine strategy





Context Optimization

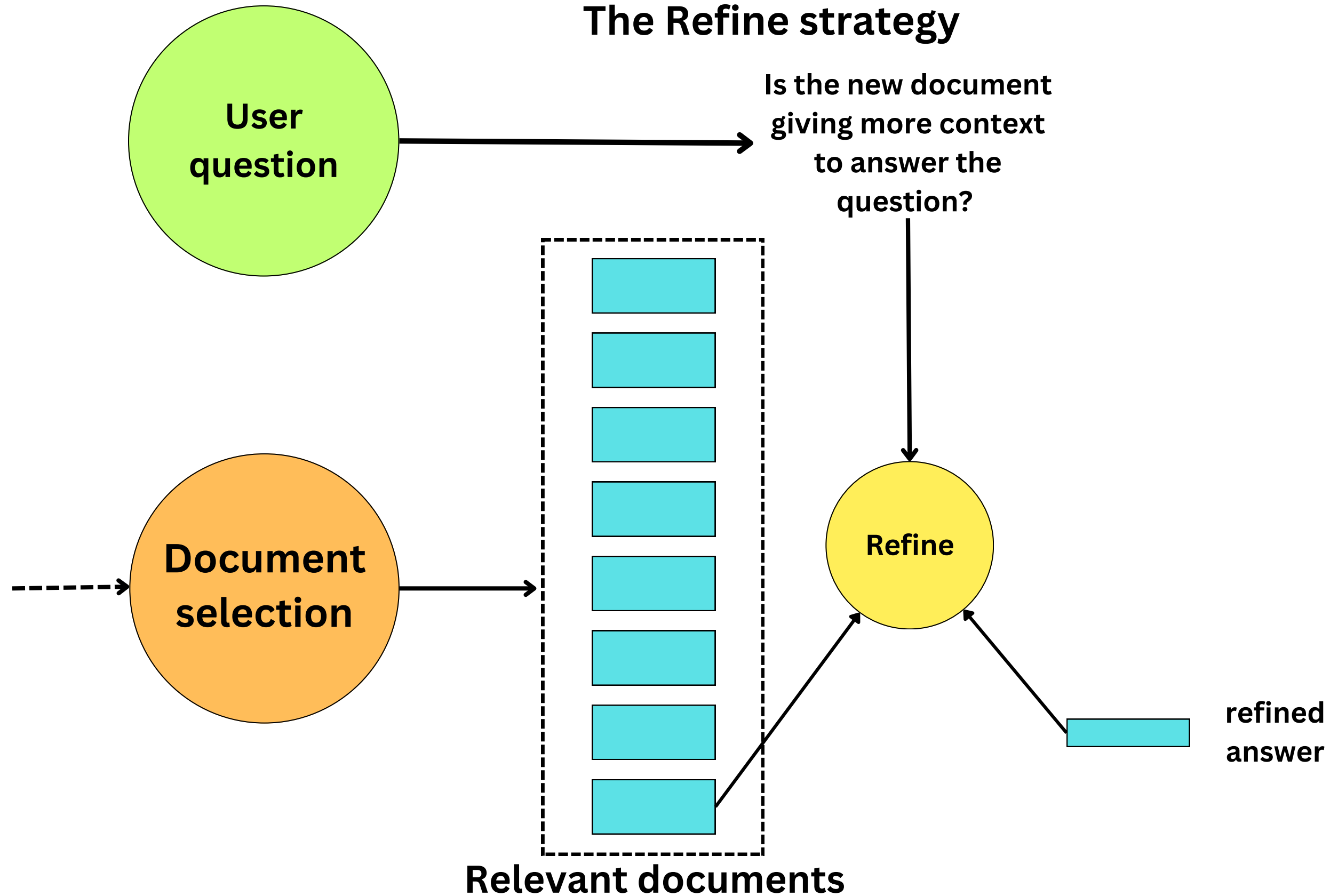
The Refine strategy





Context Optimization

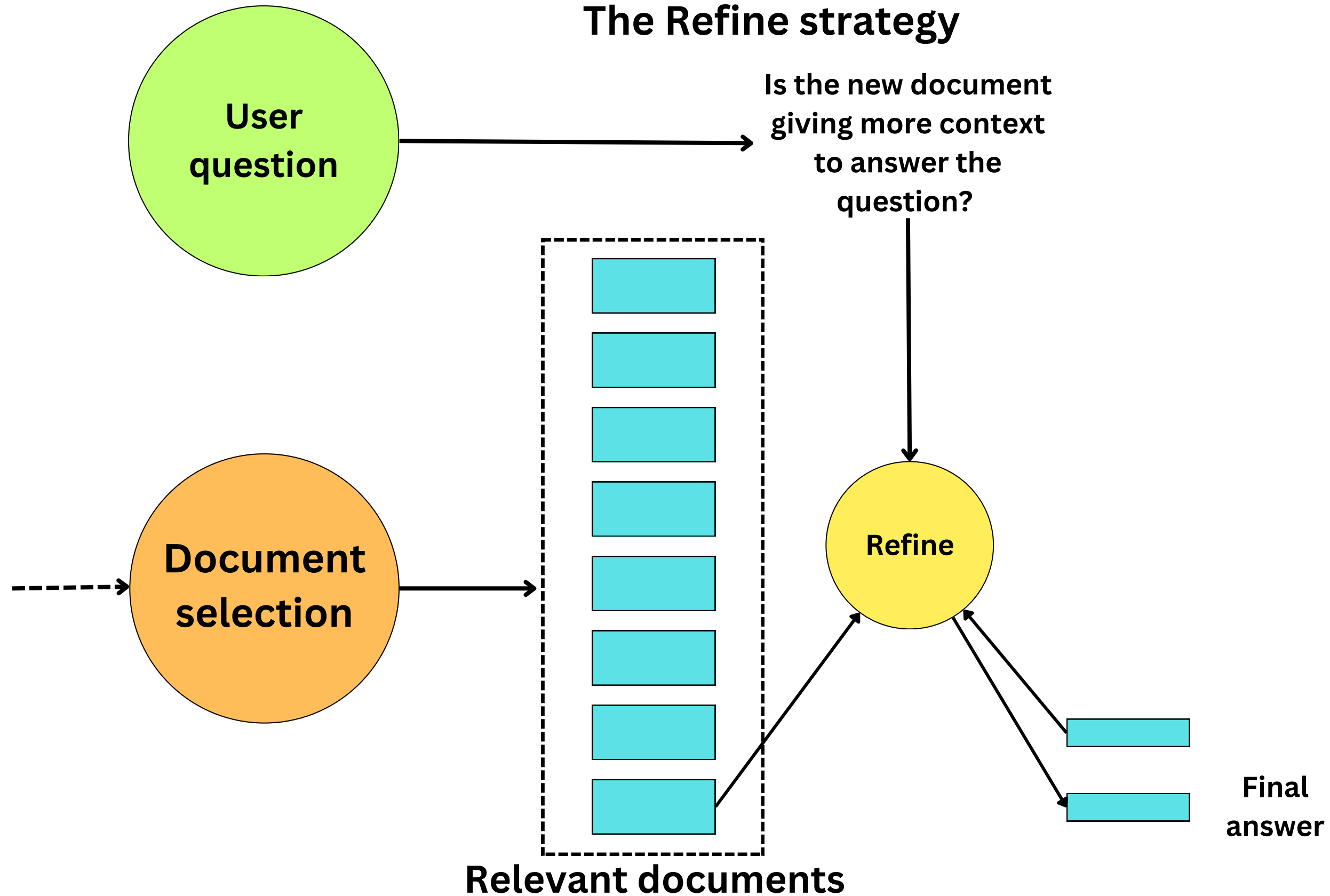
The Refine strategy





Context Optimization

The Refine strategy





Context Optimization

The Refine strategy

